Trees and Ensembles

Lesson 7 — Technologies for Artificial Intelligence

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Lesson plan

Time	Minutes	Segment	Material
0:00–0:10	10	Exercise 6 discussed; the choice this lesson offers	Slides 2–5
0:10–0:32	22	Decision trees: splitting on Gini impurity	Slides 6–20

Time	Minutes	Segment	Material
0:32–0:52	20	Depth is the bias-variance dial	Slides 21–27
0:52–1:12	20	**Notebook 01**, or atoms 1–7 — decision trees from scratch	Slide 28
1:12–1:24	12	**Break**	Slide 29
1:24–1:49	25	Bagging and random forests	Slides 30–48
1:49–2:02	13	**Notebook 02**, or atoms 8–13 — bagging and random forests	Slide 49
2:02–2:24	22	Gradient boosting	Slides 50–60
2:24–2:42	18	**Notebook 03**, or atom 14 — gradient boosting	Slide 61
2:42–3:00	18	The full leaderboard; homework	Slides 62–67
	180	**Total**	**67 slides, 3 notebooks**

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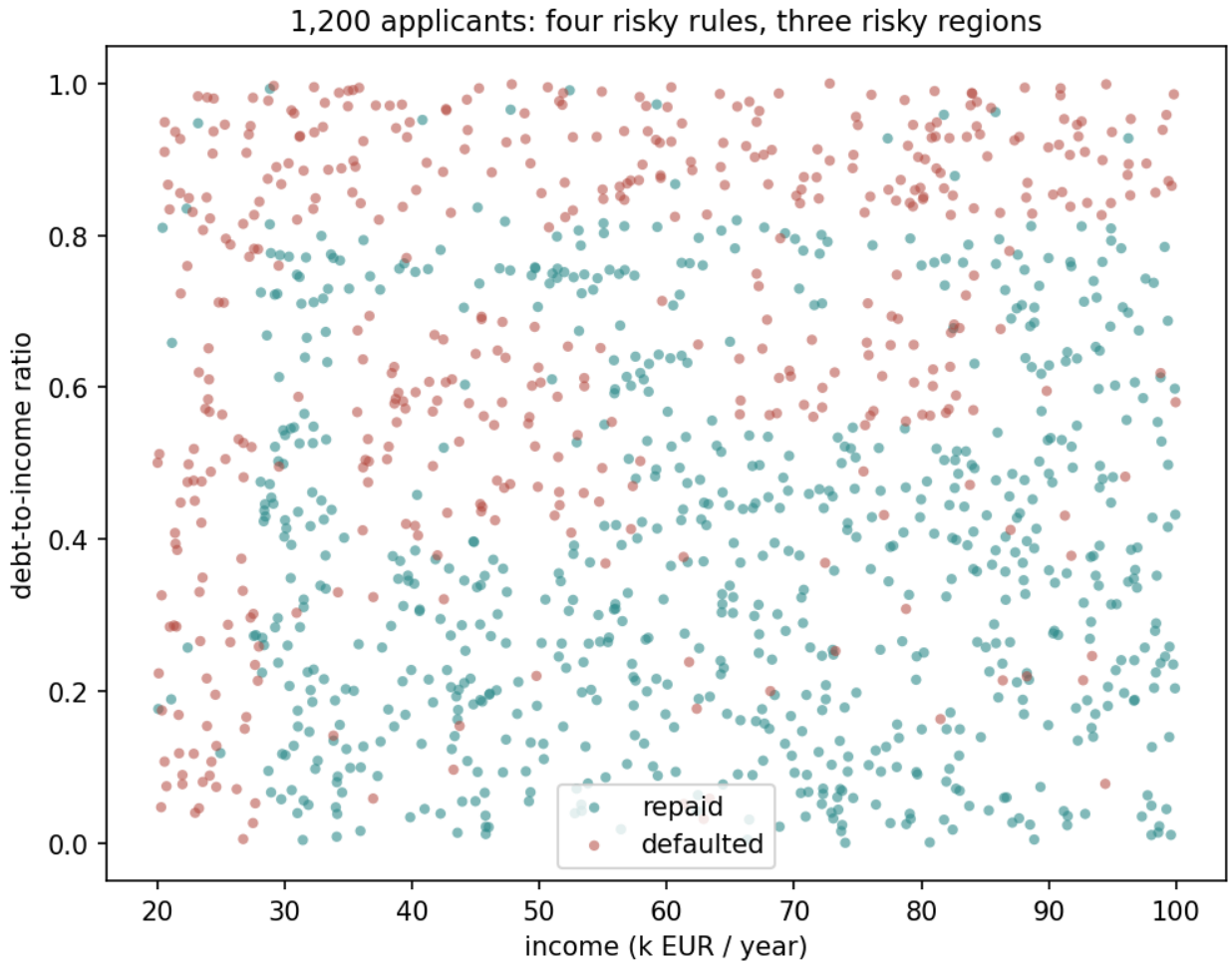
1. Why this lesson exists

Lessons 3 and 4 fit a straight line and made it honest. Lesson 6 offered three ways of bending, remembering or assuming your way around a boundary a line cannot draw. This lesson introduces a fourth family, built on a completely different primitive — a threshold, not a distance or an inner product — and then spends most of its time on the question that actually decides how well trees work in practice: what happens when you build more than one.

A single decision tree is one of the few methods in this course a person without a mathematics background can read end to end. It is also, on its own, a mediocre model: accurate enough to be tempting, unstable enough to be untrustworthy. **Ensembles of trees — bagging, random forests, gradient boosting — are the method most competition leaderboards on tabular data are still won with**, precisely because they turn that instability into their raw material rather than trying to eliminate it.

1.1 The dataset, and the shape a line cannot cover

1,200 loan applicants, two figures each at the point of application: income in thousands of euros a year, and debt-to-income ratio. Whether the loan later defaulted is not a straight function of either. A bank's underwriting logic is closer to a checklist than a formula: an income floor below which almost nobody is approved regardless of anything else, a debt ceiling above which almost nobody is, and two "stressed" combinations in between — income and debt levels that are each unremarkable alone but risky together.



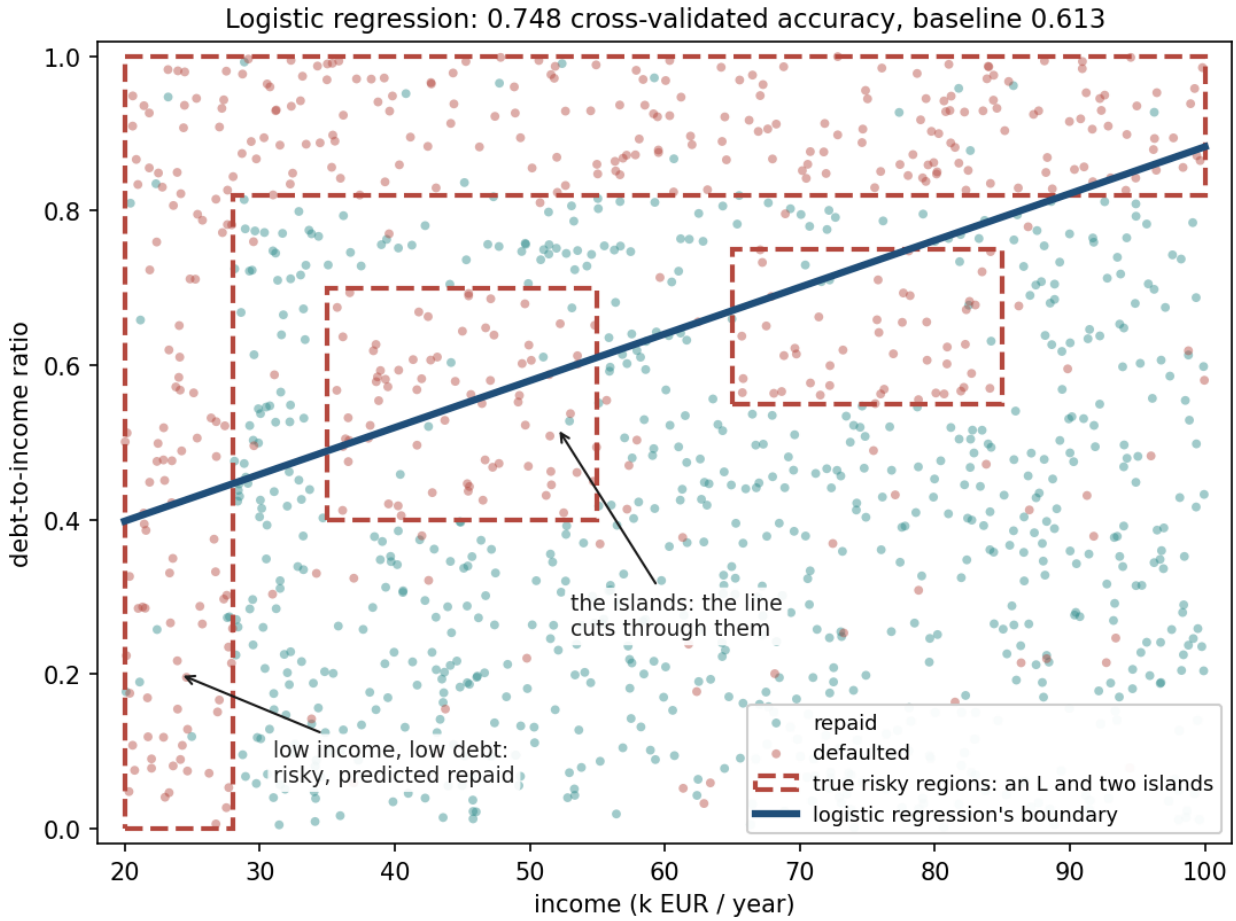
1,200 applicants. Four rules make an applicant risky: low income, or high debt, or one of two rectangles where income and debt are each ordinary but the combination is not. The low-income strip (left edge) and the high-debt strip (top) meet in the top-left corner and form one L-shaped region; the two rectangles stand apart from it. Four rules, three separate regions. No single straight line separates repaid from defaulted, because the class you would have to isolate is not even one connected region.

A predictable mistake is to read the floor and the ceiling as "low income *and* high debt", the top-left corner only. The reading is reasonable, because the two stressed rectangles really are an *and* of two ranges. But the floor and the ceiling each apply on their own, so the risky region they make is the whole L: an applicant with income below the floor is risky at any debt, and one with debt above the ceiling is risky at any income.

7% of labels are flipped after the rule is applied — the noise floor this lesson works against, and the reason no model below should be read as failing when it falls short of 1.000. The ceiling of 0.93 is an average: each label is flipped with probability 7%, and in this particular sample 85 of the 1,200 were, so the true rule itself scores 1,115 / 1,200 = 0.929 on the observed labels. A cross-validated score can land a little either side of 0.93 by chance. 38.7% of applicants defaulted (464 of 1,200), so the majority baseline is 0.613.

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression	**0.748 ± 0.030**

Unlike lesson 6's pump data — where logistic regression matched the baseline exactly and had learned nothing — here it clears the baseline by a wide margin. Two of the four rules, the income floor and the debt ceiling, are monotonic in one feature — the risk rises steadily as income falls or as debt rises — and a linear boundary captures monotonic effects reasonably well. Where it goes wrong is easiest to see by drawing the line it fits.



The boundary logistic regression draws when fitted on all 1,200 applicants: above the blue line it predicts default, below it repaid. The dashed outlines are the true risky regions. Look at the three places the line disagrees with them: the bottom of the low-income strip, below the line; the right end of the high-debt strip, where the line climbs above the ceiling; and the two islands, which the line cuts straight through.

The line runs from a debt ratio of 0.40 at an income of 20 thousand euros to 0.88 at 100 thousand: almost a threshold on debt alone, tilted slightly. Its three failures are the three places the true rule is not a half-plane. Of the 87 applicants with income below the floor and debt below the ceiling, it predicts 50 as repaid, though the rule makes all of them risky. At high income it misses debt ratios between the ceiling, 0.82, and the line, 0.88. And it cannot wall off the two *interior* islands without

calling part of each healthy and the healthy territory beside them risky. A boundary built from thresholds, applied more than once, can do all three, and that is what the rest of this lesson builds.

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2. Decision trees: splitting on Gini impurity

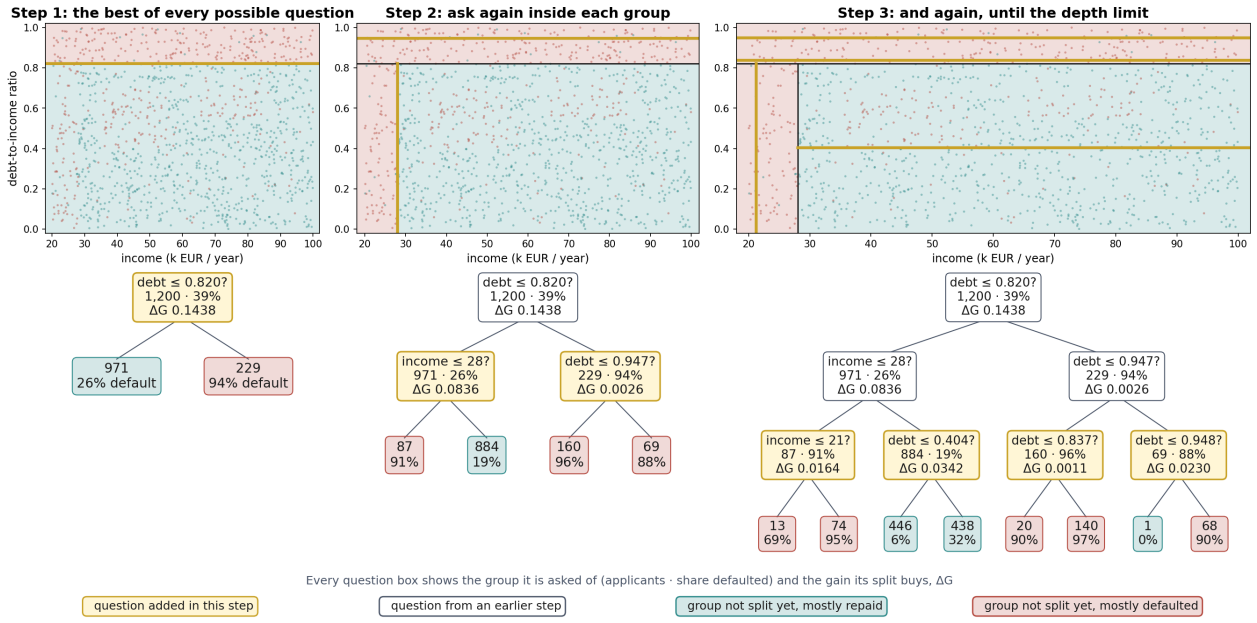
2.1 The whole algorithm

> Find the single (feature, threshold) split that makes the two resulting groups as pure as possible. Apply it. Repeat on each group, independently, until a stopping rule is met.

Nothing is estimated in the sense lessons 3 and 4 used the word — no coefficient, no gradient with respect to a parameter vector. The tree still learns from the data which feature to ask about and where to put each threshold, but it learns them by trying candidates and keeping the best, not by descending a slope. A tree is a sequence of yes/no questions, chosen greedily, and "greedily" matters: at every step the tree takes whichever split helps most *right now*, with no mechanism for looking ahead to a split that would help more two levels down.

A predictable mistake is to read the algorithm as "one feature per level, applied to all the examples, then the next feature". It is a natural reading: a drawn tree looks like a fixed sequence of questions, and the first question really is asked of everyone. It is wrong twice over. Nobody chooses the feature: at every node the tree tries every feature and every threshold and keeps the best pair. And after the first split each group is searched on its own, so the two groups can be asked different questions, and a feature can come back as often as it helps — in the figure below, one path asks about debt three times.

The recursion is easiest to see run. Three rounds of it on the loan data:



Each column is one more round of "find the best question, apply it, repeat inside each group". On top, the cuts in the plane of the data, the ones added in that round in gold; underneath, the tree asking the same questions. Every box shows its applicants and their default rate, a question as well as a group not yet split: when a group is split its box becomes the question asked of it, so the 971

of step 1 reads "income 28?" in step 2, with 971 under it, and splits into 87 + 884. Step 2 asks a different question in each group, because each is searched on its own. Step 3 shows greed: one cut, at 0.948 — a thousandth above the one at 0.947, so the two gold lines overlap — sets aside a single applicant from 68, another splits 160 into 90% and 97% default and changes no prediction, and neither interior island has been found yet. Each question box also shows the gain its split buys, ΔG (section 2.2): in step 2, 0.0836 for the income question and 0.0026 for the debt one, whose two groups are both still mostly defaulted, so that split changes no prediction either.

How little the later rounds buy is worth counting. The four splits of step 3 change exactly one prediction in 1,200: training accuracy is 0.8400 at depth 2 and 0.8408 at depth 3. And that one applicant is a flipped label: income 23.22, below the floor, and debt 0.9475, above the ceiling, so both rules say default, yet the label says repaid. The tree spent a split memorising one noisy label, which is section 3's story about depth in miniature. It splits anyway because it judges impurity, not decisions: any split whose ΔG is above zero is taken.

The stopping rules act branch by branch. A group that is pure, or too small to split, or at the depth limit, stops its own branch; the other branches carry on, and the tree is finished when every branch has stopped. Without a depth limit every branch runs until its group is pure.

The loop, written out. Everything in this section and the next, in the order the code runs it (notebook 1's `build_tree` and `best_split`):

```
def grow(g):
  if stop(g): return leaf(g)
  gain = {c: ΔG(g, c) for c in splits(g)}
  best = max(gain, key=gain.get)
  if gain[best] <= 0: return leaf(g)
  L, R = split(g, best)
  return node(best, grow(L), grow(R))
```

Line by line: `g` is the group being grown; `stop(g)` is true when the group sits at the depth limit, is pure, or has fewer than 2 examples; `leaf(g)` predicts the group's majority class; `splits(g)` lists every candidate, a feature with one of its midpoint thresholds; `ΔG(g, c)` is the gain of section 2.2 on the two groups candidate `c` would make; `max(gain, key=gain.get)` picks the candidate with the largest gain; `split` makes the two groups, `L` and `R`; `node` keeps the question and its two subtrees.

ΔG, defined in section 2.2, does two jobs, both at one node at a time: it ranks that node's candidates, and it decides whether the node is split at all. Each child then runs the whole loop again on its own examples, with its own impurity in the parent's place and its own size as m; gains are never added up across nodes or compared between them in this loop (section 4's `max_leaf_nodes` is the exception, and compares them on purpose).

What the two constructors return is worth saying, because the code hides it. `leaf(g)` is a new object that keeps one thing, the majority class of `g`, as its prediction; it does not keep the examples. `node(best, grow(L), grow(R))` holds the question and the two subtrees. Python evaluates the arguments first, so both recursive calls run all the way down before the node that holds them is built: the tree is assembled from the leaves upward, depth first. At the root, `node((debt, 0.82), grow(the 971), grow(the 229))`. And `max(gain, key=gain.get)` returns the candidate with the largest gain, not the gain itself. **Two predictable mistakes** come from reading it too fast: "the function returns `g`" (it returns a leaf built from `g`) and "`max` gives the best gain" (it gives the best candidate).

Training builds the tree; inference walks it. To predict for a new applicant, start at the root; at each question go left if the applicant's value is at or below the threshold, right otherwise, until a leaf, and return its class (notebook 1's `predict_one`). The two costs are lopsided: training scores every candidate at every node, while a prediction is at most `max_depth` comparisons, 8 for the best tree here, however large the training set was. Lesson 6's k-nearest neighbours is the opposite: nothing to train, and every prediction compares the new point with all the training points.

2.2 Impurity, and the gain a split buys

"As pure as possible" needs a number. For classification, the standard choice is the **Gini impurity** of a group of m_g examples drawn from C classes:

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

where p_c is the fraction of the group belonging to class c. The classes are the labels a group can hold: here $C = 2$, defaulted and repaid, and the two shares add up to 1. Slide 8's right-hand group of step 1, before it is split again, holds 229 applicants of whom 215 defaulted, so $p = 215/229 = 0.939$ for defaulted and $14/229 = 0.061$ for repaid, and $G = 1 - (0.939^2 + 0.061^2) = 1 - 0.885 = 0.115$: nearly pure. For two classes this simplifies to $2p(1-p)$: expand $(1-p)^2$ as $1 - 2p + p^2$, and the two p^2 terms and the two 1s cancel, leaving $G = 1 - p^2 - (1 - 2p + p^2) = 2p - 2p^2$. It is 0 when the group is pure ($p \in \{0, 1\}$) and maximal, 0.5, at $p = 0.5$. **A predictable mistake:** "Gini never exceeds 0.5." Reasonable, since every group in this lesson has two classes. But 0.5 is the two-class maximum; with C equally frequent classes $G = 1 - C \times (1/C)^2 = 1 - 1/C$, so three classes reach 0.667.

What that number is, exactly. G is not a probability of misclassification. It is the probability that two examples drawn at random *with replacement* from the group, and labelled according to the group's own class frequencies, would **disagree** — and the formula is that statement written out. Draw twice: the two draws agree on class c with probability p_c^2, so they agree at all with probability $\sum_c p_c^2$, and they disagree with the rest:

$$P(\text{disagree}) = 1 - \sum_{c=1}^{C} p_c^2 = G$$

A pure group cannot disagree with itself, which is why $G = 0$ there.

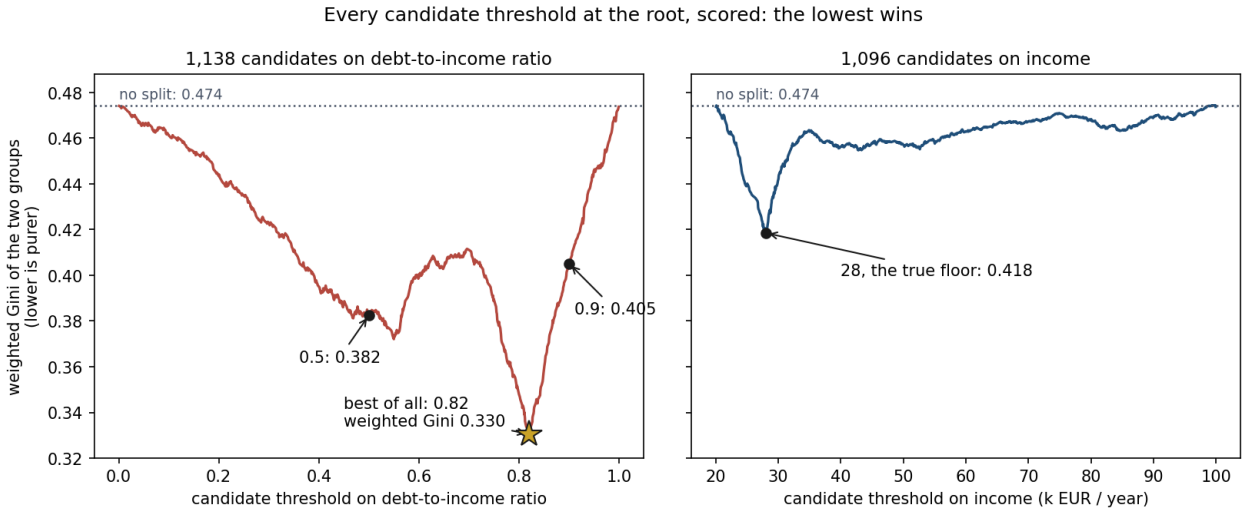
A note of history. The name is Italian. Corrado Gini (1884–1965) was a statistician and the first president, in 1926, of the Istituto Centrale di Statistica, the body that became ISTAT. $1 - \sum_c p_c^2$ is his index of heterogeneity ("mutabilità", 1912); Breiman, Friedman, Olshen and Stone made it the splitting criterion of their Classification and Regression Trees (CART) in 1984, and scikit-learn kept it. It is not the Gini coefficient of income inequality, his better-known index — the same man, a different quantity.

A candidate split divides a parent group of m_P examples into a left child of m_L examples and a right child of $m_R = m_P - m_L$. Its value is the reduction in impurity, weighted by how the group is divided:

$$\Delta G = G(\text{parent}) - \left(\frac{m_L}{m_P} G(\text{left}) + \frac{m_R}{m_P} G(\text{right}) \right)$$

For a continuous feature, the tree scans every feature and, within each, every midpoint between two adjacent sorted values as a candidate threshold, and keeps whichever (feature, threshold) pair maximises ΔG. It then repeats the entire search independently inside each child. This is what makes the algorithm greedy: ΔG is evaluated one split at a time, never for a pair of splits jointly, so a feature that only helps in combination with a second split further down can be — and in this dataset's two interior islands, is — invisible to a shallow tree.

Infinitely many thresholds, finitely many splits. A threshold is a real number, so there are infinitely many to choose from, but they make only finitely many different splits: any two thresholds between the same two neighbouring values send exactly the same examples left. At the root, two neighbouring debt ratios in the data are 0.8198 and 0.8205; a threshold of 0.8199, 0.8200 or 0.8204 makes the same 971 / 229 split. One threshold per gap is therefore enough, and the convention is the midpoint, $(0.8198 + 0.8205)/2 = 0.82015$, which keeps it as far as possible from both values seen in training. A feature with k distinct values has $k - 1$ gaps: debt ratio has 1,139 distinct values here and income 1,097, so the root search scores $1{,}138 + 1{,}096 = 2{,}234$ candidates. The search is exact, not an approximation — the predictable mistake is to assume it must be approximate because thresholds are continuous.



Every candidate threshold at the root, scored by the weighted Gini impurity of the two groups it makes (lower is purer; the dotted line, 0.474, is no split at all). Look for the lowest point on either curve: debt at 0.82, 0.330. Debt at 0.5 leaves both groups mixed (0.382); income's best threshold, near the true floor of 28, only reaches 0.418.

A predictable mistake is to read "purest" as "the most defaults on one side". The score counts proportions, each weighted by its group's size, not numbers of defaults. Debt at 0.9 makes a right-hand group almost as pure as 0.82 does — 129 applicants at 92% default against 229 at 94% — but leaves the large left group more mixed, 1,071 at 32% against 971 at 26%, and its weighted impurity is 0.405 against 0.330. A group pure in "repaid" counts as much as one pure in "default": $2p(1-p)$ does not ask which class is common.

Worked example. On the full 1,200 applicants, $p = 0.387$ defaulted, so $G(\text{parent}) = 2 \times 0.387 \times$

$0.613 = 0.474$. The best root split found by exhaustive search is `debt_ratio <= 0.82` — recognisably the debt ceiling the data was built with, though the tree does not know that rule: it takes the midpoint of two neighbouring debt values in the data, 0.8198 and 0.8205, which rounds to the same 0.82 — and the from-scratch implementation below confirms this is exactly the split scikit-learn's `DecisionTreeClassifier` also finds. More than that: at every depth tested the two make the same prediction for every one of the 1,200 applicants. The check runs on the training data on purpose — it tests the code, whether two programs make the same decisions from the same rows, not how well the tree will do on applicants it has not seen. Identical predictions do not by themselves prove the two trees are the same node by node, since two different trees can agree on every row they are shown; the notebook therefore also prints the root split from both, and it is the same.

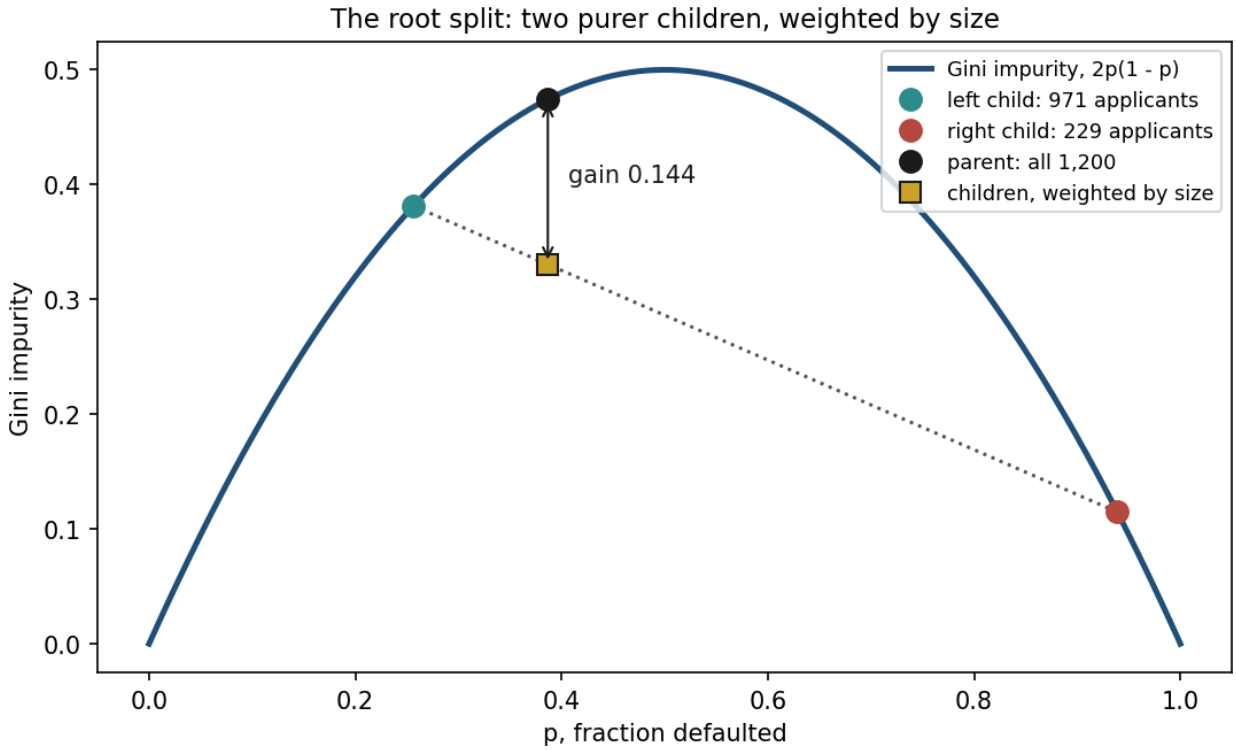
What that split buys, in the formula above. It sends 971 applicants left and 229 right. The rule first: count the defaults in each group, divide by its size to get p, and put p into $2p(1-p)$:

Group	Applicants m	Defaulted	p = defaulted / m	$G = 2p(1-p)$
Parent	1,200	464	464 / 1,200 = 0.387	$2 \times 0.387 \times 0.613 = 0.4743$
Left, `debt_ratio <= 0.82`	971	249	249 / 971 = 0.256	$2 \times 0.256 \times 0.744 = 0.3814$
Right	229	215	215 / 229 = 0.939	$2 \times 0.939 \times 0.061 = 0.1148$

The defaults add up, $249 + 215 = 464$, as they must. (The G column is printed from the unrounded p, which is why its fourth decimal can differ from the product of the rounded values shown.) The two children are weighted by their share of the parent, 971/1,200 = 0.809 and 229/1,200 = 0.191: $0.809 \times 0.3814 + 0.191 \times 0.1148 = 0.3086 + 0.0219 = 0.3305$, below the parent's 0.4743. In one line:

$$\Delta G = 0.4743 - \left(\tfrac{971}{1200} \times 0.3814 + \tfrac{229}{1200} \times 0.1148\right) = 0.4743 - 0.3305 = 0.1438$$

Two predictable mistakes in this arithmetic. The first is to subtract the weighted average from 1 instead of from the parent's impurity: $1 - 0.3305 = 0.6695$, larger than the parent's whole impurity. It is natural, because the Gini formula itself starts with "1−"; the check is that a split cannot remove more impurity than the parent had, so $\Delta G \leq G(\text{parent})$, and with two classes never above 0.5. The second is to rebuild a count from a rounded percentage: the figure in section 2.1 prints the left group as 26% default, and $0.26 \times 971 = 252$, not 249 — the true share is $249/971 = 25.6\%$. The counts come first.



The Gini curve $2p(1-p)$ with the parent (black) and the two children (teal and rust). The children's size-weighted average (gold) lies on the straight line between them, and the curve bulges above every such line — so any split that separates the classes at all lowers the weighted impurity. The gap, 0.144, is the gain; the tree takes the split whose gap is largest. The square sits exactly under the parent because the children's p, averaged with the same weights, gives the parent's back — with counts it is exact, $\frac{971}{1200} \cdot \frac{249}{971} + \frac{229}{1200} \cdot \frac{215}{229} = \frac{249+215}{1200} = \frac{464}{1200} = 0.387$ — because a split moves defaulters between groups, it does not create any.

> **Try this:** using `Notebooks/loan_data.py`, compute G for the group of applicants with `income_k < 28` (the income-floor rule alone) and compare it with G for the full dataset. It should be substantially lower — that subgroup is disproportionately defaults — which is exactly why the tree is willing to spend a split isolating it.

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3. Depth is the bias-variance dial

Lesson 6 turned k in k-nearest neighbours and watched bias trade directly against variance. A tree's equivalent dial is **depth**: how many questions any one applicant may be asked on the way from the root to a leaf, where the tree stops and predicts a class by majority vote. Depth counts questions along one path, not in the whole tree: a tree of depth 2 can ask three questions in all, one at the root and one in each of its two branches.

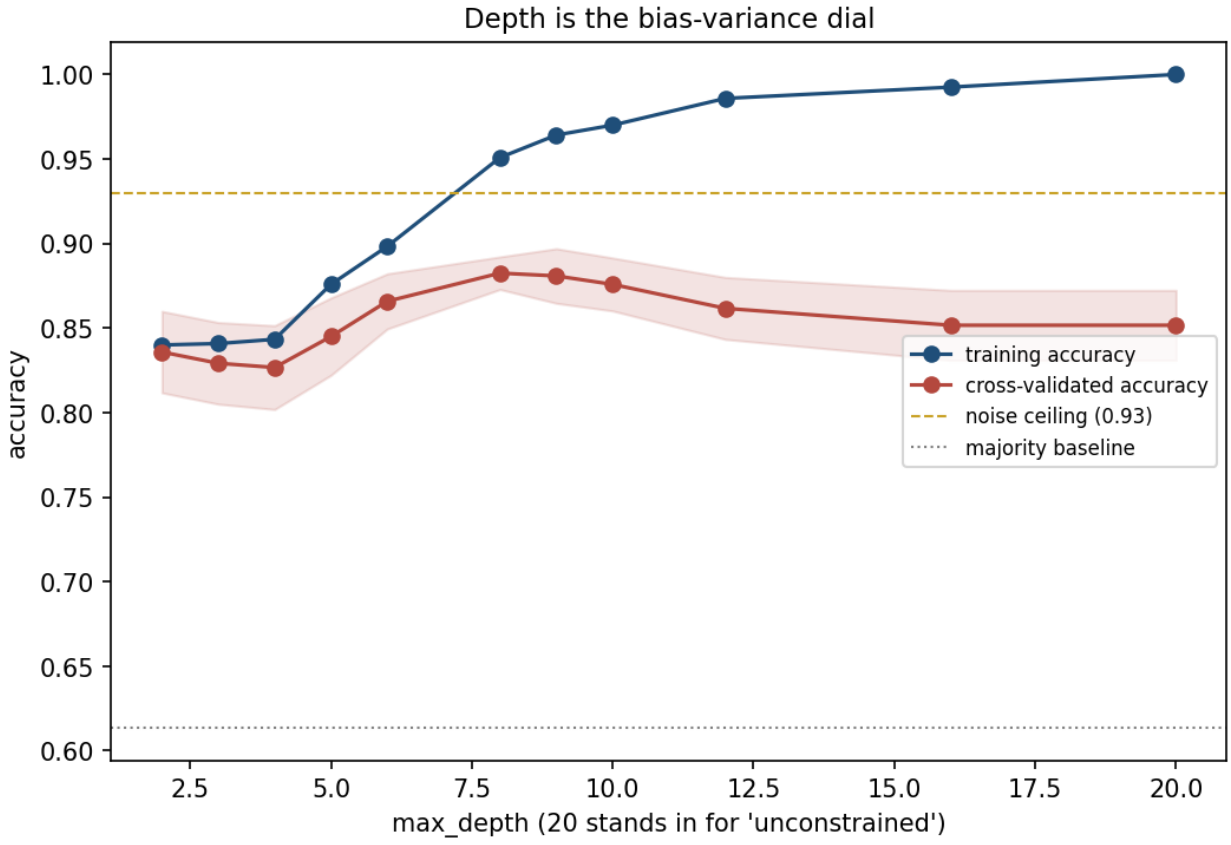
Shallow tree. Few questions, so only the coarsest structure can be described. High bias, low variance.

Deep tree. Enough questions to isolate almost any subset of the training data, down to single points. At full depth, training accuracy is **exactly 1.000**, even on a dataset with no structure at

all: keep splitting and every leaf is eventually pure, at worst one example, which it then "predicts" perfectly. The one exception is two examples with identical feature values and different labels, which no split can separate; with two continuous features drawn at random, as here, that does not occur. That number demonstrates the model class is flexible enough to memorise, not that anything has been learned — the same reading lesson 6 gave $k = 1$'s training accuracy of 1.000.

`max_depth`	Training accuracy	Cross-validated, 5 folds (mean ± standard deviation)
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
9	0.964	0.881 ± 0.016
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

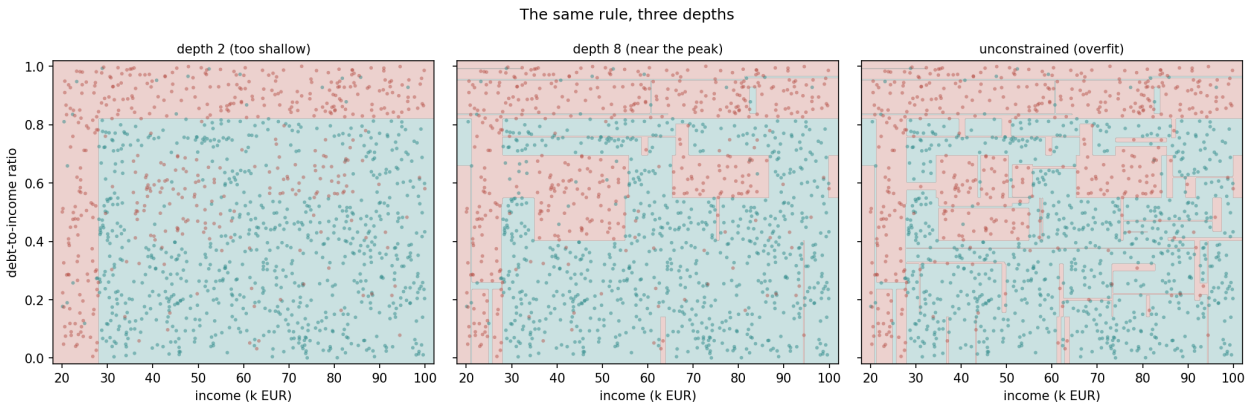
The cross-validated column is lesson 5's five-fold stratified cross-validation: the 1,200 are cut into five folds of 240 with the same default share, and five times the tree trains on 960 and is scored on the 240 it did not see. At depth 8 the five scores are 210, 211, 215, 214 and 209 correct out of 240, so the mean is $(0.875 + 0.879 + 0.896 + 0.892 + 0.871)/5 = 0.882$, and the ± is their standard deviation, 0.010: how much the score moves from one fold to another, not a confidence interval.



Training accuracy climbs monotonically towards 1.000. Cross-validated accuracy rises, peaks at depth 8, and then falls back — even though the model keeps getting more flexible. Past the peak, more and more of each extra split is spent fitting the 7% of labels this dataset deliberately flips, and a flipped label has no pattern to learn; a tree that classifies it "correctly" on the training set has memorised it, not generalised. The shaded band is ± one standard deviation across the five folds; "20" on the axis stands for the unconstrained tree, whose real depth is 24.

Two ways to see the overfitting. *Who votes in a leaf.* A leaf predicts the majority of its applicants. At depth 2 the tree has 4 leaves with a median of 123.5 applicants each, a vote of a hundred; the unconstrained tree has 156 leaves with a median of 2, and 72 of them hold a single applicant — a prediction made from one person's label, lesson 6's $k = 1$. *The line at 0.929.* The true rule itself scores 1,115/1,200 = 0.929 on these labels, because 85 were flipped. A training accuracy above that line means the tree agrees with the labels better than the truth does, so it must be reproducing flipped labels: depth 8 is just past it at 0.951, and the unconstrained tree has learned all 85. The ceiling (the gold line) is what a model can expect, on average, on *new* applicants, whose flipped labels it cannot know; on its own training data it can do far better by memorising, which is why the training curve crosses the line and the cross-validated one does not. **A predictable mistake** is to read a training score above the ceiling as a bug. On training data it is expected. On new data 0.93 is an average, not a wall: the number of flipped labels in a test set varies, so even the rule that generated the data scores above 0.93 on about half of all possible sets of 1,200 new applicants (it needs 83 flips or fewer, where 84 are expected: probability 0.484). A score a little above the ceiling on new data can be luck; one far above it, or above it on every split, is a bug.

Read the two columns against each other, never in isolation. The peak sits at depth 8 — deep enough to isolate both stressed islands, shallow enough to leave the flipped labels as errors rather than as leaves of their own. The unconstrained tree **loses 3 points of cross-validated accuracy relative to the peak** while gaining nothing but a training score nobody should have trusted in the first place.



The same rule at three depths. At depth 2 every applicant answers at most two questions. The tree asks three in all — it is step 2 of the figure in section 2.1 — but the third, on the high-debt side, splits 229 applicants into two groups both predicted "default", so it changes no prediction and draws no line. The boundary shows only the income floor and the debt ceiling — the two biggest, easiest-to-find regions — and neither interior island is visible at all. At depth 8 both islands appear, close to the rectangles of the generating rule though not exact, and a few thin slivers have already started — its training accuracy, 0.951, is past the 0.929 line. At full depth the boundary has grown a fringe of tiny rectangles chasing individual flipped labels, in both colours — rust in healthy territory,

teal inside the risky strips: not a smoother version of the depth-8 boundary, a noisier one.

The predictable mistake, and why the instinct is sound. It is entirely reasonable to expect a more flexible model to do at least as well as a less flexible one on held-out data — more options can only help, surely. The instinct is sound for *bias*: a deeper tree can represent anything a shallower one can, plus more. It is wrong for the quantity that is actually measured, because flexibility increases variance at the same time, and past some point the added variance costs more than the reduced bias saves. The course has shown this curve bend before — lesson 3's test error turned back up as the polynomial degree grew, lesson 6's as k shrank towards 1 — and it will not be the last time.

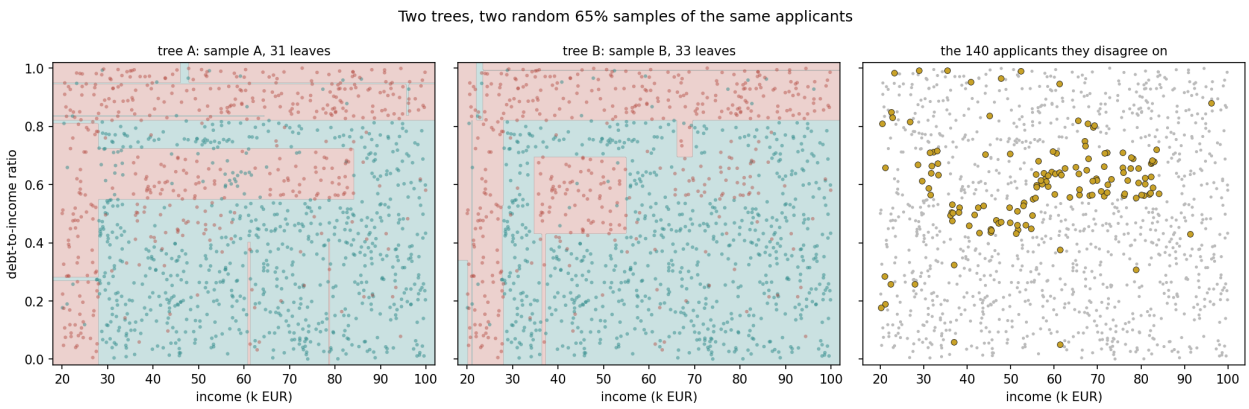
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4. Reading a tree, and why it doesn't stay still

The two knobs scikit-learn offers limit different things. `max_depth` is a **cap on every path**: each node above the cap is split if it can be, whether or not that particular split is worth having — the depth-2 tree of section 3 spent one of its three questions on a split that changed no prediction. `max_leaf_nodes` is a **budget of leaves** for the whole tree, spent **best-first**: the tree always expands whichever current leaf offers the largest gain — compared across all the current leaves, each leaf's ΔG weighted by its share of all the examples — so a small budget of splits goes to the ones that matter most. It is the one place where gains at different nodes are compared. For a tree meant to be read rather than merely scored, that is the better knob.

With `max_leaf_nodes = 9` the tree reaches 90.8% training accuracy in nine leaves and, more to the point, **0.901 cross-validated** ($\pm$ 0.020) — above the depth-8 tree's 0.882 with its 71 leaves. Nine leaves spent where they gain most beat 71 spent evenly, because this dataset's rule is made of axis-aligned rectangles, the very shape a few well-placed thresholds draw. (The 9 was chosen for readability, not by trying budgets on the folds.) Every threshold is recognisable from the generating rule: 28 for the income floor, 0.82 for the debt ceiling, 34.92 and 58.00 tracing the outline of the first stressed island. That printout is the entire model — a sequence of if/else statements a loan officer could apply by hand — and it is the thing k-nearest neighbours and support vector machines cannot offer at all. For a decision that has to be justified to the person it affects, that can matter as much as accuracy; this lesson's companion reading takes the point further.

The cost is **instability**. An early split near the root reroutes everything beneath it, so a small change in the training data can change the tree's shape. Two trees at `max_depth = 6`, each fit to its own random 780 of the 1,200 applicants — drawn **without replacement**, so no one twice, like cards dealt without putting them back; the two samples share 499 — find the debt ceiling within 0.001 of each other: the evidence for it is overwhelming, so it barely moves. Inside each sample the ceiling separates 636 applicants at about 25% default from 144 at 94% (24.5% and 24.8%, 93.8% in both). The two thresholds, 0.82015 and 0.81935, differ only because each is the midpoint between the two neighbouring debt values its sample holds, as in section 2.2: sample A has applicants at 0.8198 and 0.8205; sample B lacks the one at 0.8198, so its neighbours are 0.8182 and 0.8205. But they grow 31 and 33 leaves and give **opposite verdicts to 140 of the 1,200 applicants** (11.7%): for those 140, the answer depends on which sample the bank happened to train on.



The two trees' boundaries and, right, the 140 applicants they disagree on. Look at where the gold points sit: three in four in or right next to the two islands (105 of 140 within 4 thousand euros of income and 0.05 of debt ratio of them, 75 inside), where each split was decided by a few dozen applicants; of the other 35, 19 lie in the two strips (11 below the income floor, 11 above the debt ceiling, 3 of them in the corner where both apply) and 16 elsewhere. The root split is stable because the signal behind it is unambiguous and every applicant of the sample is behind it; the splits below it, each supported by far fewer points, are not, and they are most of the tree: 31 leaves mean 30 questions, one at the root and 29 below it.

A predictable mistake: "two trees disagree on 11.7% of the applicants, so each tree is wrong on 11.7%". Reasonable, since a disagreement and an error both mean a verdict that differs from something. But a disagreement compares the two trees with each other, not with the labels: on each of the 140 applicants exactly one of the two is right, and elsewhere both can be right or both wrong. Disagreement measures instability; accuracy needs the labels.

A single tree's instability looks like a flaw. It is the raw material sections 5–8 turn into a strength.

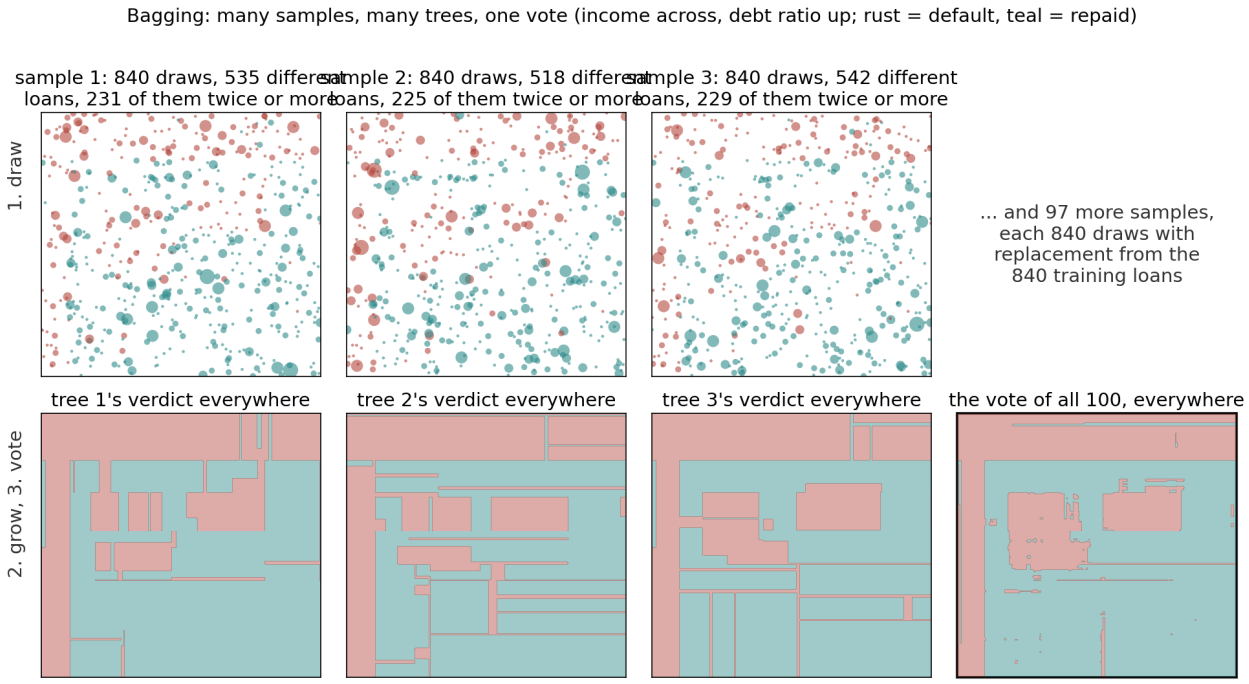
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5. Bagging: averaging away the instability

Section 4 ended with two trees that give opposite verdicts to 140 of the 1,200 applicants. If a tree's mistakes are somewhat random — different across resamples of the same data — then many trees, each grown on its own resample and combined by a vote, should have mistakes that partly cancel. That is **bagging** (**b**ootstrap **agg**regat**ing**):

> Draw a bootstrap sample: m rows chosen **with replacement** from the m training rows, so some appear more than once and some not at all. Fit a tree to it — in this lesson a fully grown one, though bagging itself works with any tree, or any model. Repeat `n_estimators` times. To predict, ask every tree and take the majority vote.

A note of history: bagging is Leo Breiman's, published in 1996 as "Bagging predictors" — the same Breiman who co-wrote the CART trees of 1984 and, five years later, the random forest.



Bagging drawn, on the 840 training loans of notebook 2's first split. Top: three of the 100 bootstrap samples — a dot's size grows with the times that loan was drawn, and missed loans are absent. Bottom: the fully grown tree each sample produces, three different shapes from the same 840 loans, and the majority vote of all 100, which keeps the strips and the islands and drops most single-tree slivers.

Nothing about the tree itself changes — bagging reuses section 2's algorithm unmodified and changes only what surrounds it. Three details are worth stating. The m rows are the **training** rows: in notebook 2, the 840 of a 70/30 split. Drawing m of them with replacement, rather than a smaller share without, gives every tree as many rows as a single tree would have, while still making every sample different. And the trees are independent of one another: each needs only its own sample, so they can be grown in parallel (`n_jobs` in scikit-learn). scikit-learn's default is `n_estimators=10`; notebook 2 uses 100.

One detail of scikit-learn's `BaggingClassifier`: it does not count votes, it averages the trees' predicted probabilities. With fully grown trees the two are the same thing. Every leaf is pure, so each tree's probability is 0 or 1, and averaging 0s and 1s is counting votes. Notebook 2's own majority-vote version and scikit-learn's therefore apply the same rule, though not to the same trees: each draws its own bootstrap samples. They still make the same prediction for all 360 applicants of its test split, 0.872 accuracy both — with 100 trees, the luck of any one bootstrap sample is outvoted.

Why a vote cancels a tree's mistakes, one applicant at a time. A tree's verdict on an applicant depends on two things: what the data really say in that part of the plane, the same for every tree, and the luck of its own bootstrap sample, different for every tree. A mistake caused by luck falls on different applicants for different trees, so on any one applicant the unlucky trees are a minority and the vote outvotes them. In notebook 2's test split, the applicant at income 39.22 and debt ratio 0.9721, who defaulted, gets "default" from 69 of the 100 trees and "repaid" from 31, tree 1 among them; the vote is right. Over the 360 test applicants, one tree alone is wrong on 58.7 on

average; the vote gets 18.9 of those right, and is itself wrong on 46.

So bagging does not *create* disagreements: a single tree already has them, unseen. The bootstrap imitates collecting the loans again, without new data, and **bagging averages what an unstable tree would have answered on many plausible datasets**. It follows that bagging helps only unstable models. On the same split, a depth-2 tree scores 0.806 and a bagging of 100 depth-2 trees also 0.806: those trees barely change from one sample to the next, and there is nothing to average. The unconstrained tree goes from 0.825 to 0.872.

A comparison worth making: is bagging to a deep tree what cross-validation is to logistic regression? No. Cross-validation *measures* how much a result depends on the sample; bagging *reduces* that dependence. The closer analogy is lesson 3's Ridge on the degree-12 polynomial: a flexible, unstable model and a remedy for its variance — Ridge by restraining the coefficients, bagging by averaging many models. Neither touches the bias.

A predictable mistake, and why it is reasonable. "The trees cancel each other's errors." Section 4's picture is all disagreement, so the reading is natural. But the vote cancels only the part of the mistakes that changes from tree to tree. A mistake every tree makes the same way stays. The flipped labels are the clearest case: on a new applicant whose label was flipped, the trees have learned the rule that generated the data from its neighbours, and they vote what the rule says — on average 95 trees in 100. Of the 29 flipped labels among notebook 2's 360 test applicants, the vote gets 1 right. On the training applicants it is the other way: 55 of the 56 flipped labels come out "right", because about two trees in three (67%) hold the row in their sample and have memorised it. Either way the trees agree, and there is nothing to cancel. The flip is random, independent of income and debt, so no model could predict it except by luck — it is what the noise ceiling of 0.929 leaves out.

5.1 How much data one bootstrap sample actually sees

One draw picks a given row with probability $1/m$, so it misses it with probability $1 - 1/m$. The m draws are independent, so the chance that all of them miss it is

$$\left(1 - \frac{1}{m}\right)^m$$

As $m \to \infty$, this is the standard exponential limit $\lim_{m\to\infty} \left(1 + \frac{x}{m}\right)^m = e^x$ with $x = -1$:

$$\left(1 - \frac{1}{m}\right)^m \xrightarrow{m\to\infty} e^{-1} \approx 0.368$$

so each bootstrap sample contains about $1 - e^{-1} \approx 63.2\%$ of the distinct rows and leaves about **36.8%** out entirely — called **out-of-bag** (OOB). At $m = 1{,}200$ the finite-sample value is $(1 - 1/1200)^{1200} = 0.3677$, already within 0.0002 of the limit, and a single simulated bootstrap draw left out 36.6% of rows — close to both, with the residual gap being exactly the sampling noise a single draw carries.

On average, then, a sample of 1,200 draws holds $1{,}200 \times (1 - 0.3677) = 758.7$, about **759**, distinct rows, and the other $1{,}200 \times 0.3677 = 441.3$, about **441**, draws are repeats.

A predictable mistake, and why it is reasonable. "So each tree is trained on 63.2% of the data, about 759 rows." The number is the one just derived, so the reading is natural. But a bootstrap

sample still has $m = 1{,}200$ draws: about 759 *different* rows, and the other 441 draws are repeats of rows already drawn. The tree is fit to 1,200 rows, some of them two or three times over, which is what the next paragraph counts.

One identity follows from the counting. The 1,200 draws are the different rows plus the repeated draws; the 1,200 rows are the different rows plus the rows never drawn. So the repeated draws are exactly as many as the out-of-bag rows: every repeat takes the place of a row left out. The rows drawn twice or more are fewer, about $1{,}200 \times 0.264 = 317$, because some take three or four draws. In notebook 2 the bagging is trained on 840 rows, so a sample holds about $840 \times 0.632 = 531$ different loans.

A second predictable mistake. "The out-of-bag rows are left out of the training set." The word *out* invites it, because it sounds like a test set. But out-of-bag is decided tree by tree: a row is missing from *this* tree's sample, and the other trees, about 63% of them, do see it. It is a training row like any other. That is exactly what makes out-of-bag validation free, below.

How often is a row drawn exactly r times? The same counting answers it. A row is drawn exactly r times when r of the m draws pick it and the other $m - r$ miss it. There are $\binom{m}{r}$ ways to choose which r draws pick it; each of those r picks it with probability $1/m$, and each of the other $m - r$ misses it with probability $1 - 1/m$:

$$P(r) = \binom{m}{r} \left(\frac{1}{m}\right)^r \left(1 - \frac{1}{m}\right)^{m-r}$$

For large m two things happen. First, $\binom{m}{r}/m^r = \frac{m(m-1)\cdots(m-r+1)}{r!\,m^r}$ tends to $1/r!$, because each of the r factors $m, m-1, \ldots$ divided by m tends to 1. Second, $(1 - 1/m)^{m-r}$ tends to e^{-1}, as above. So

$$P(r) \xrightarrow{m\to\infty} \frac{e^{-1}}{r!}$$

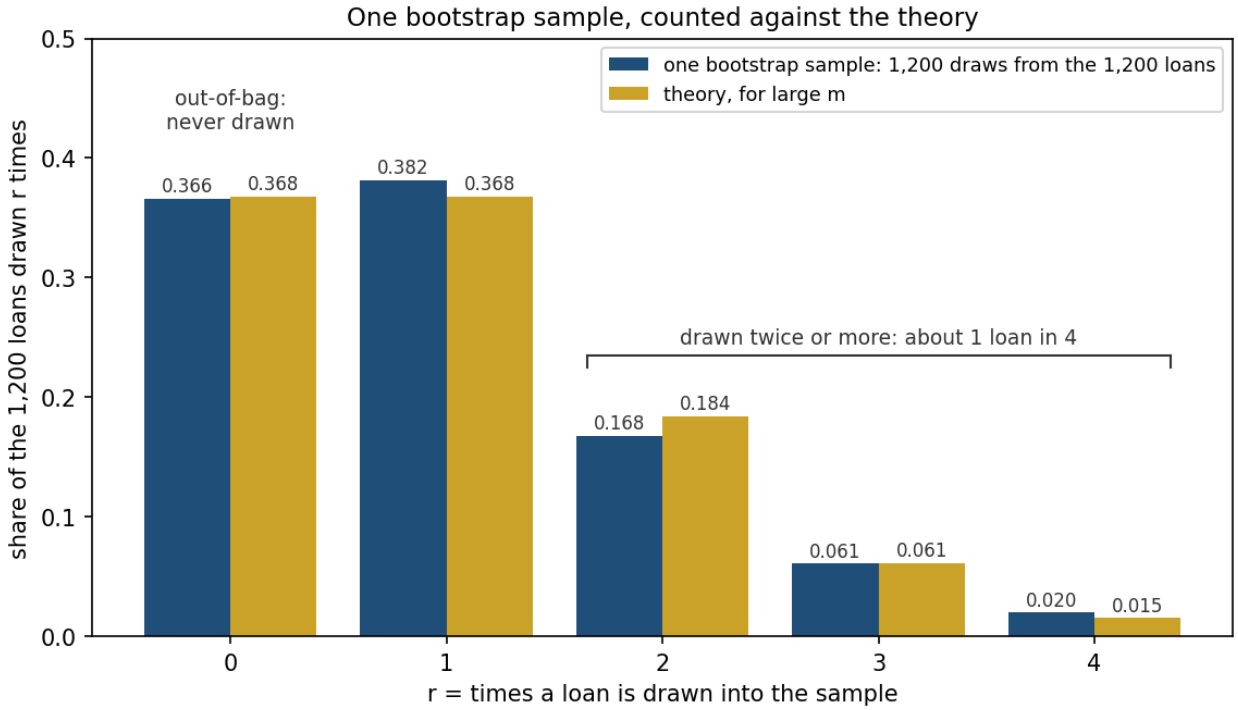
Why "once" is as common as "never" has a picture: throw 1,200 balls at random into 1,200 boxes, one ball per box on average, and there are about as many empty boxes as boxes with exactly one ball. In three steps. *Never*: every one of the 1,200 draws must miss the row, $(1 - 1/1{,}200)^{1{,}200} = 0.3677$. *Exactly once*: draw number 1 picks it and the other 1,199 miss it, $(1/1{,}200)(1 - 1/1{,}200)^{1{,}199}$; but the draw that picks it can be any of the 1,200, so there are 1,200 such cases, and $1{,}200 \times 1/1{,}200 = 1$, leaving $(1 - 1/1{,}200)^{1{,}199} = 0.3680$. *The comparison*: the two formulas differ only in the exponent, one factor $1 - 1/1{,}200 = 0.9992$, so $P(1)/P(0) = 1{,}200/1{,}199$. The reason is that each row is drawn once on average; with 2,400 draws, "once" would be twice as likely as "never".

In words: $r = 0$ leaves nothing to choose, so $P(0) = e^{-1}$, the out-of-bag share. $r = 1$ has m choices, each worth $1/m$, so $P(1) = 1 \times e^{-1}$ too — a row is drawn once exactly as often as never. $r = 2$ has $m(m-1)/2$ pairs, each worth $1/m^2$, which is about $\frac{1}{2}$ (at $m = 1{,}200$, $\binom{1200}{2}/1200^2 = 0.4996$), so $P(2) \approx e^{-1}/2$. This is the **Poisson distribution with mean 1**, the limit of m draws each with probability $1/m$.

A note of history: Siméon Denis Poisson published it in 1837, in *Recherches sur la probabilité des jugements en matière criminelle et en matière civile* — a study of how often juries convict, where it appears as the limit of many trials each with a small chance.

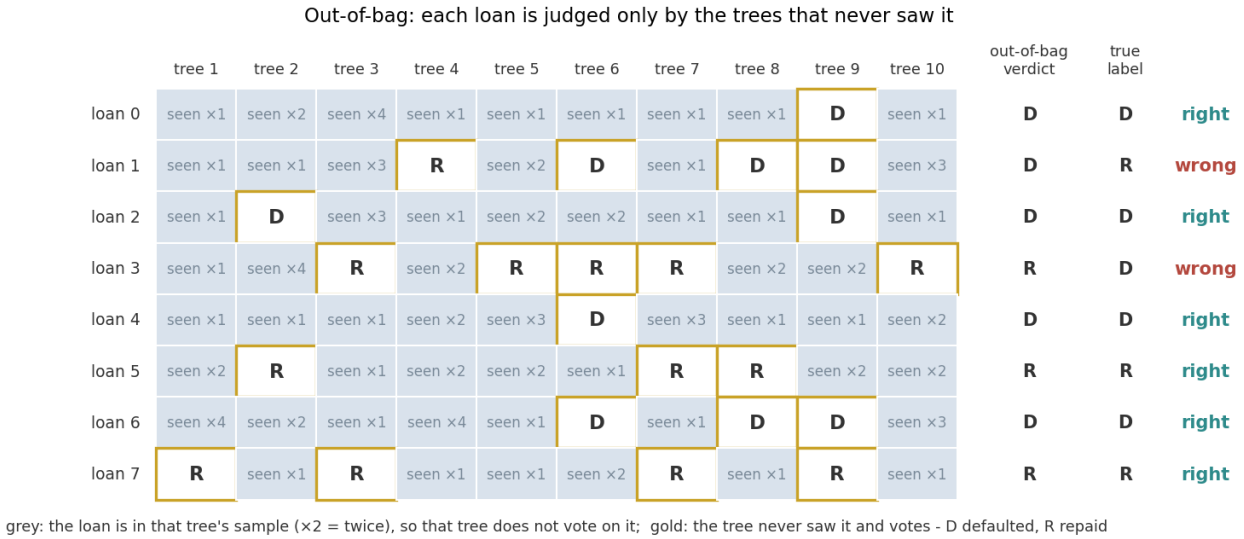
r	Exact at $m = 1{,}200$: $\binom{m}{r}(1/m)^r(1 - 1/m)^{m-r}$	Limit: $e^{-1}/r!$	One sample (notebook 2): loans drawn r times $\div$ 1,200
0	0.3677	$0.3679 \div 1 = 0.3679$	0.366
1	0.3680	$0.3679 \div 1 = 0.3679$	0.382
2	0.1840	$0.3679 \div 2 = 0.1839$	0.168
3	0.0613	$0.3679 \div 6 = 0.0613$	0.061
4	0.0153	$0.3679 \div 24 = 0.0153$	0.020

At $m = 1{,}200$ the exact shares and the limit agree to the fourth decimal, and one real sample sits close to both. Drawn twice or more: in theory $1 - 2e^{-1} = 0.264$ of the rows; in the sample, 303 of the 1,200 — about one row in four.



Each pair of bars is one value of r, the number of times a loan is drawn into the sample; a bar's height is the share of the 1,200 loans drawn exactly r times — blue in one real sample, gold the theory for large m. The $r = 0$ bars are the out-of-bag loans. About one loan in four is drawn two or more times, and that mix of repeats and absences is what makes each tree different from its neighbour.

Out-of-bag: free validation. Every out-of-bag row is unseen data for the trees that missed it. So judge each row by a vote of only those trees, compare the verdict with its label, and count over all rows: that share is the **OOB score**, with no rows set aside and no extra fit.



Ten bagged trees on all 1,200 loans, the first eight loans shown. Grey: the loan is in that tree's sample (×2 = drawn twice), so that tree does not vote on it. Gold: the tree never saw the loan, and its vote is shown, D for defaulted, R for repaid. The out-of-bag verdict is the majority of the gold votes (an exact tie goes to "repaid", the first class, as in scikit-learn); the OOB score is the share of loans, out of all 1,200, where it matches the true label. With only ten trees a loan can sit in every sample and have no gold vote at all: the chance is $0.632^{10} = 0.0102$*, about 12 of the 1,200 loans. That is why the real computation, next, uses 300 trees.*

At full size, notebook 2 fits 300 bagged trees on all 1,200 loans, with no split. Each loan is judged by the trees that never saw it — 110.5 of the 300 on average, as $0.368 \times 300 = 110$ predicts, from 81 to 133 — and the vote is right on 1,090 of the 1,200 loans: an OOB score of **0.9083**. To compare, 5-fold cross-validation of the same ensemble trains five ensembles, each on 960 loans, and scores each on the other 240: 212, 215, 223, 222 and 211 right, 1,083 of 1,200, a mean of **0.9025 ± 0.021**. Both are accuracies, the share of verdicts that are right. Step by step:

Round: the group left out	Right of its 240	Accuracy = right ÷ 240	Deviation from the mean of the five = accuracy − 0.9025	Deviation²
1	212	0.8833	−0.0192	0.000367
2	215	0.8958	−0.0067	0.000044
3	223	0.9292	+0.0267	0.000711
4	222	0.9250	+0.0225	0.000506
5	211	0.8792	−0.0233	0.000544

The mean of the five accuracies is their sum divided by 5, 0.9025 — equal to 1,083/1,200 because the groups are the same size. The standard deviation is $\sqrt{0.002174/5} = 0.0208$. numpy's `np.std` divides by 5, the number of rounds; dividing by 4, as some statistics texts do, gives 0.023.

The two answer the same question — how does the ensemble do on loans it never saw? — with different bookkeeping. In 5-fold cross-validation each fold is judged by a whole ensemble trained

without it, and the work is done five times. Out of bag, each loan is judged by the trees whose bootstrap sample missed it: about a third of the ensemble, a different third for each loan. There is no single out-of-bag set. The practical consequence: with a bagged ensemble you get an honest estimate of accuracy on unseen data without setting aside a validation set at all.

Checks. The two estimates agree to well within cross-validation's own spread of ± 0.021, and the gap is luck of a small size, not a law: over twelve seeds it averages 0.0040, and the worst is 0.0075.

5.2 Why averaging reduces variance, and what it cannot fix

The idea first, in words. Each tree's verdict on an applicant depends on the luck of its bootstrap sample, and averaging many trees cancels that luck. But every tree grows from the same 1,200 loans: what those particular loans say, every tree learns, and what is shared does not cancel. It is like averaging ten experts who read the same newspaper — their personal quirks cancel, the newspaper's mistakes do not. So averaging stops at a floor, and the only way lower is trees that are less alike.

The same, in symbols. Fix one applicant. Let σ^2 be the variance of one tree's verdict on it — over the randomness of training: which loans the lender happened to collect, and which bootstrap sample the tree drew. Let ρ be the correlation between two trees' verdicts on it, over the same randomness. With the data held fixed, two bootstrap trees are independent; ρ is above zero because all the trees share one dataset, so when the data change, they move together. For B trees averaged together:

$$\operatorname{Var}\left(\frac{1}{B}\sum_{b=1}^{B} f_b\right) = \frac{1}{B^2}\left(\sum_{b=1}^{B}\operatorname{Var}(f_b) + \sum_{b\neq b'}\operatorname{Cov}(f_b, f_{b'})\right)$$

There are B variance terms, each σ^2, and $B(B-1)$ covariance terms, each $\rho\sigma^2$ — the same accounting lesson 5, Section 4.3 did for k correlated fold scores, which is not a coincidence: averaging correlated measurements is one piece of arithmetic, and this is its second appearance in the course:

$$= \frac{1}{B^2}\Big(B\sigma^2 + B(B-1)\,\rho\sigma^2\Big) = \frac{\sigma^2}{B} + \frac{B-1}{B}\,\rho\sigma^2 \xrightarrow{B\to\infty} \rho\sigma^2$$

With $\rho = 0.5$: ten trees give $0.1 + 0.9 \times 0.5 = 0.550$ of one tree's variance, a hundred give $0.01 + 0.99 \times 0.5 = 0.505$.

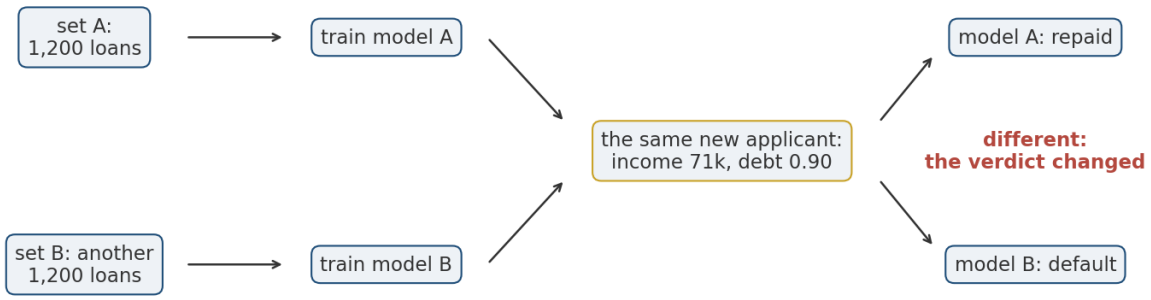
Two consequences follow directly from the algebra. First, variance falls towards a **floor** of $\rho\sigma^2$, not to zero: the private part, $(1-\rho)\sigma^2$, is averaged away, and the shared part is not. Adding trees past the point where σ^2/B is small buys almost nothing. Second, averaging **cannot repair an error every tree shares**. For an average of numbers this is exact: the mean of unbiased-but-noisy predictions is unbiased, and the mean of predictions that share a systematic error keeps that error. A vote between classifiers is not an average of numbers, and it can move a point across the boundary — which is why the worked example below sees the mean accuracy rise. But no vote can fix a mistake every voter makes, as the flipped labels of section 5 showed. Bagging is above all a variance tool. It does nothing for trees that are too shallow to represent the rule in the first place.

How can the 1,200 loans mislead every tree alike? They are one sample of the lender's applicants, and a sample's chance features cannot be told apart from the rule that generated the data (section 1.1). Island A's lower edge is found at 0.404 rather than 0.40, because no applicant happens to fall

in between, and every bagged tree draws from the same 1,200; flipped labels that happen to fall close together look like a small risky pocket, and each bootstrap sample holds about 63% of them. Only more data removes such features.

The floor can also be **measured**, because the generator can draw new datasets — as if the lender had collected its 1,200 loans in another year. The experiment, in five steps: draw a set A and a set B of 1,200 loans each; train the same model on each; ask both about the same new applicant; if one says "default" and the other "repaid", the verdict changed; repeat for 300 applicants and for every pair of 40 sets (seeds 1000 to 1039; the lesson's own is 42), and count the share of applicants whose verdict changes. It measures stability, not correctness — and it is possible only because the data are synthetic, which is exactly why the bootstrap imitates other collections by resampling the one set a real lender has.

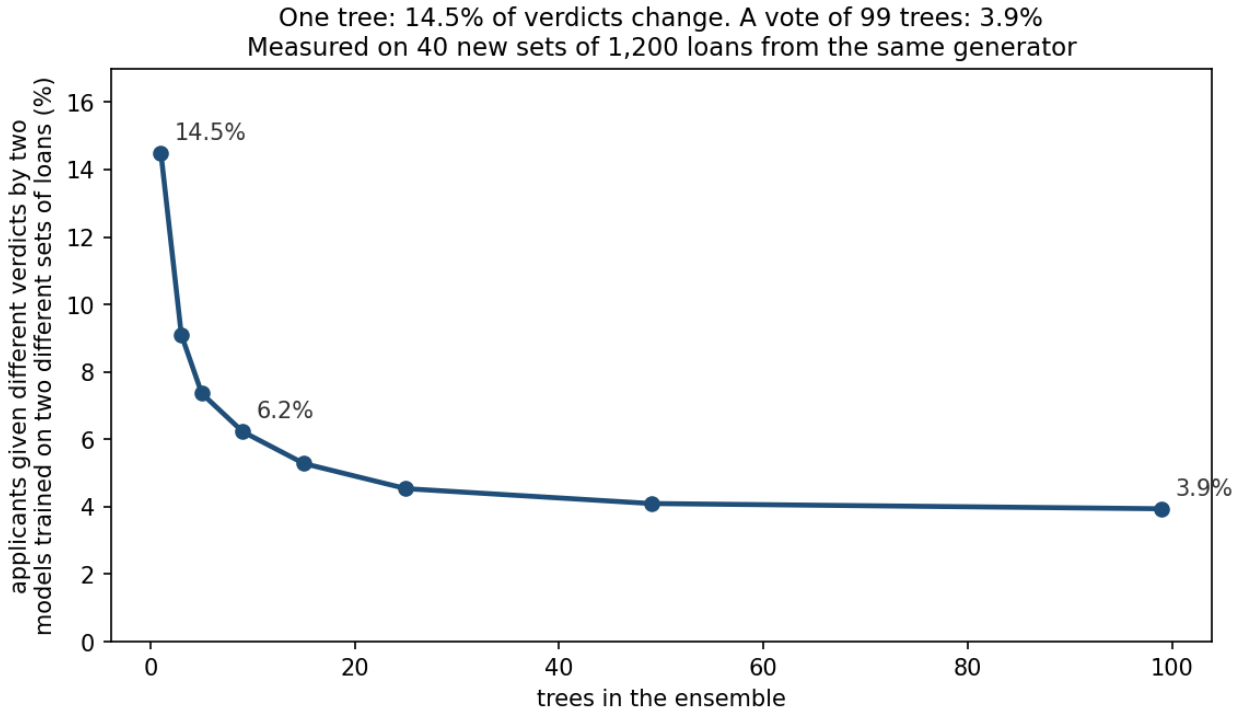
The experiment: two sets of loans, two models, one applicant



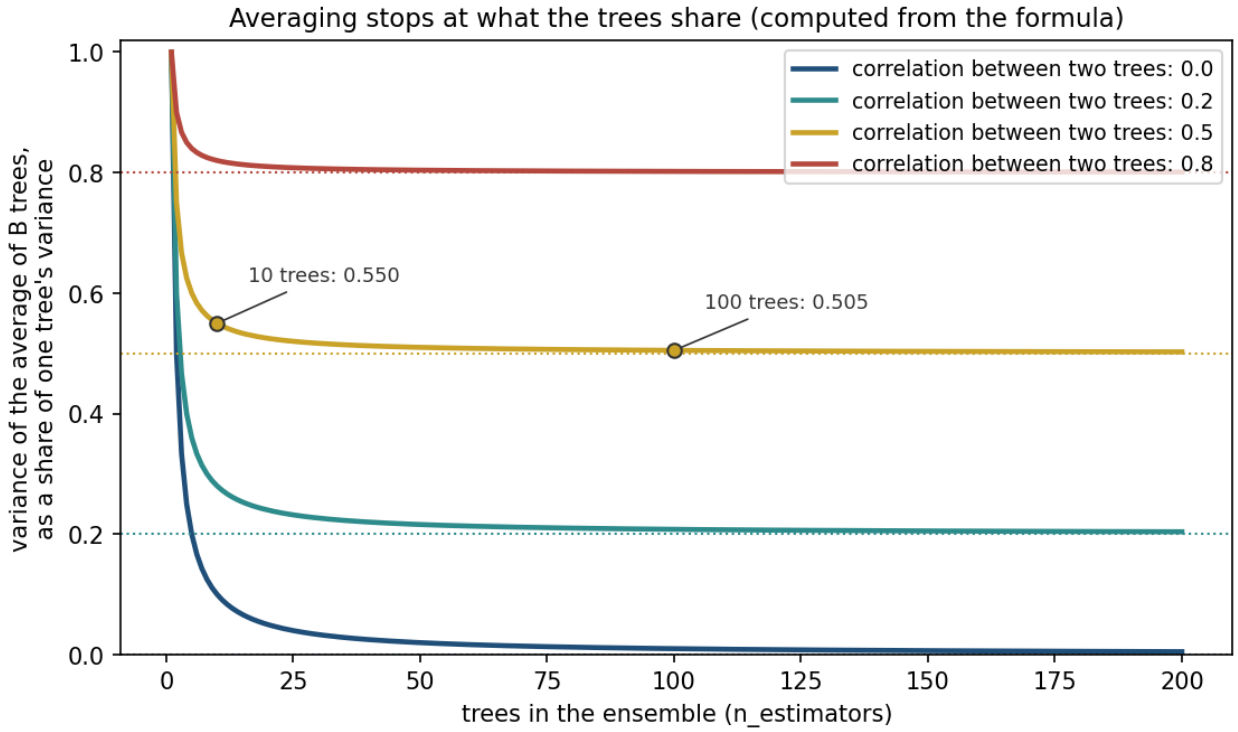
Repeat for 300 applicants and for every pair of the 40 sets of loans: on what share of applicants do the two models disagree?

The experiment, drawn, with a real applicant: at income 71k and debt ratio 0.90, one tree trained on set A says "repaid", one trained on set B says "default"; the votes of 99 trees on the two sets both say "default".

With one tree, two models trained on two different sets disagree on 14.5% of the applicants; with 9 trees on 6.2%; with 99 on 3.9%, and the curve is flat from about 50 trees on.

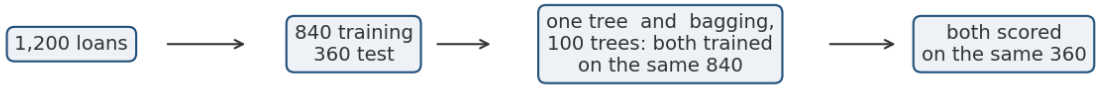


Measured on 40 new sets of 1,200 loans from the same generator: the share of applicants on which two models, trained on two different sets, give different verdicts, for an ensemble of the first 1 to 99 trees. It falls fast and stops above zero: what is left is shared by every tree, because every tree grew from the same 1,200 loans. **A predictable mistake:** "where the curve goes flat, the ensemble stops making errors". Reasonable, since a falling curve that levels off looks like a model converging. But the vertical axis counts changed verdicts between two training sets, not wrong verdicts: the floor is a floor of instability, and the ensemble is still wrong on about one applicant in ten.

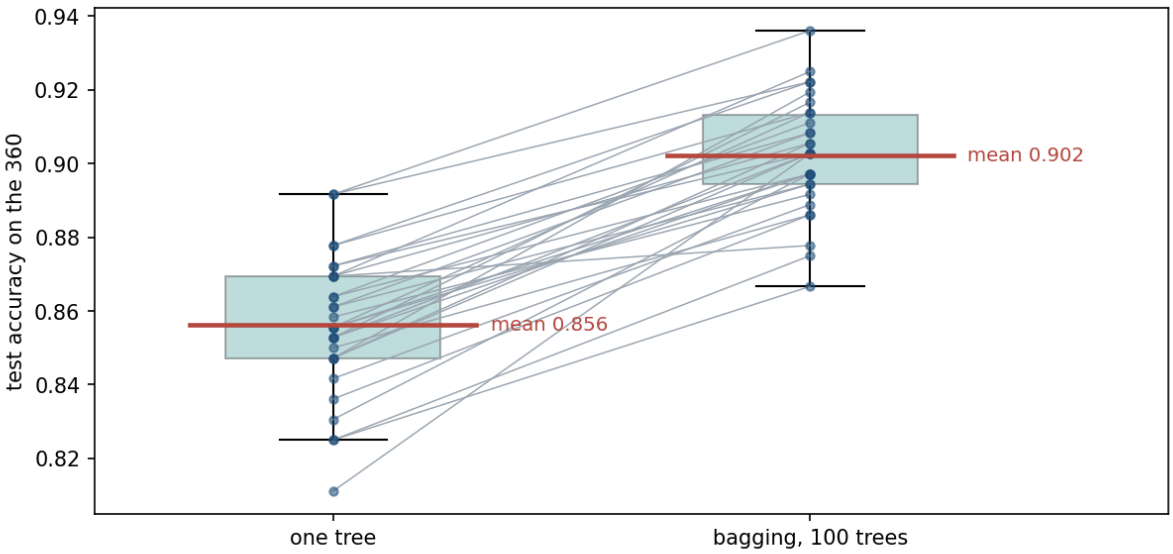


Computed from the formula above, not measured. Horizontal axis: the number of trees B. Vertical axis: the variance of the average of B trees' verdicts, as a share of one tree's variance, $1/B + \rho(B-1)/B$. One curve for each correlation ρ between two trees. Uncorrelated trees average the variance away; correlated ones stop at the dotted floor ρ however many are added — at $\rho = 0.5$, ten trees keep 0.550 and a hundred 0.505. To go lower, trees must be less alike: section 6.

Worked example. Notebook 2 splits the 1,200 loans at random into 840 for training and 360 for test, trains one unconstrained tree and a bagging of 100 trees on the same 840, scores both on the same 360, and repeats with 30 different splits.



repeated with 30 different random splits; a grey line joins the two models on the same split



box: the middle half of the 30 splits; red line: the mean; dots: the splits

The procedure on top; below, each dot one split's test accuracy, the red line the mean of the 30, and a grey line joining the two models on the same split — every line rises. The shaded box of this box plot holds the middle half of the 30 values, from the 25th to the 75th percentile, and the whiskers reach the most extreme values that are not far outliers. Bagging is higher on average and its dots a little closer together.

Model	Mean test accuracy over the 30 splits	Standard deviation across the 30
One unconstrained tree	0.856	0.0185
Bagging, 100 trees	**0.902**	**0.0153**

The comparison is paired: on each split both models see the same 840 and the same 360. Split by split, bagging is ahead on all 30, by 3 to 33 applicants.

On 360 test applicants, $(0.902 - 0.856) \times 360 = 16.6$: bagging gets about **17 more right** each time, on average, because the single tree's slivers and memorised flipped labels are outvoted. The spread is a little tighter. Much of it is the luck of which 360 applicants land in the test set, which no model removes; so this measurement is consistent with the formula rather than a direct check of it — the formula is about the variance of one prediction, and this is the spread of an accuracy over 30 different test sets.

6. Random forests: decorrelating the trees

A note of history: Breiman's "Random Forests" appeared in 2001, the same year as his essay "Statistical Modeling: The Two Cultures", a defence of judging models by how well they predict.

The idea first. Think of a committee of experts who all look at the strongest clue first: they reach the same verdicts and make the same mistakes, and averaging them is like asking one. Force each of them, at every decision, to choose among a few clues drawn at random, and their reasoning diverges. Each is a little worse alone; the committee is better, because a crowd is wise only when its members err independently.

The variance floor of section 5.2 is $\rho\sigma^2$, so going lower means making the trees less alike. Bootstrap resampling alone does this only partly, because bagged trees still consider every feature at every split, so a strong feature pulls most bootstrap trees towards splitting on it first regardless of which rows they drew.

A **random forest** is bagging — a bootstrap sample per tree, a majority vote — with one restriction: at each split, only a random subset of `max_features` features — by default $\lfloor\sqrt{n}\rfloor$ for classification — is even offered as a candidate, and the draw is made again at every split. Different trees are forced to consider different features at the same point in the tree, without any change to the splitting rule itself. It is bagging with the menu of candidate splits shrunk at random, every time a split is made.

A predictable mistake: "n is the number of classes". Reasonable, because here both are 2. But n counts the columns: with the 22 columns of section 7, each split gets $\lfloor\sqrt{22}\rfloor = 4$ of them, while the classes are still 2.

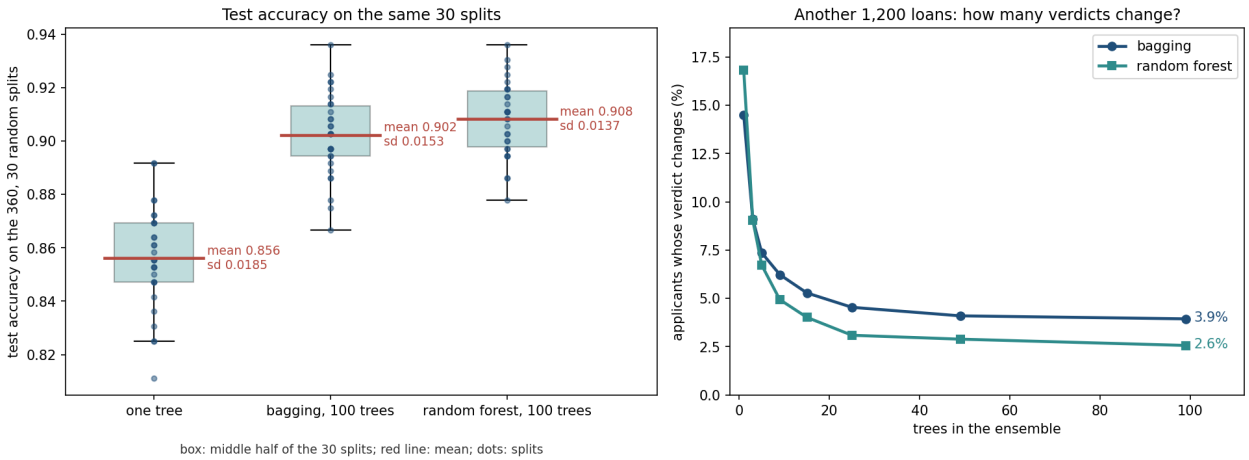
On the loan data the menu is as small as it gets. With $n = 2$, $\lfloor\sqrt{2}\rfloor = 1$: each split is offered one of the two columns, drawn at random. Notebook 2, section 4, counts what that does on the first of the 30 splits. All 100 bagged trees open with the same question, on `debt_ratio`; the forest's 100 open 51 on `debt_ratio` and 49 on `income_k`, because half the time the root is offered income alone. A tree denied `debt_ratio` at the root can still use it further down, since the draw is repeated at every split. The trees start differently, grow differently, and make their mistakes in different places. Counted the way section 4 counted two trees, two bagged trees give opposite verdicts on 38 of the 360 test applicants on average, two forest trees on 50. A wording slip worth avoiding: the forest has more *diversity* between its trees, not more variability — each tree alone is less steady, and the vote is steadier.



Three bagged trees (top: the first three) and three forest trees (bottom: the first that opens on debt and the first two that open on income, chosen to show both), with the question at their root, trained on the first of the 30 splits. The bagged trees all open on debt and draw the same strips; the forest's fill the plane differently. Right: where each ensemble's own 100 trees disagree with each other — not where bagging and the forest disagree — darker where the vote is closer to half and half; wider for the forest.

The price is paid tree by tree: denied the better column at some splits, a forest tree alone scores 0.828 mean test accuracy against a bagged tree's 0.837. The ensemble is still better, because what averaging needs is trees that err *differently*, not trees that are each the best available. **A predictable mistake** is to expect the restriction to make the forest worse, since every tree is weaker. The reasoning is right about the trees; the formula of section 5.2 says why it is wrong about their average.

Measured on the same 30 splits as section 5.2, a 100-tree random forest reaches mean accuracy **0.908** with standard deviation **0.0137** — both a better mean and a tighter spread than plain bagging's 0.902 and 0.0153, consistent with the trees being less alike. Measured as in section 5.2, on 40 fresh datasets: with another 1,200 loans the vote of 99 bagged trees changes its verdict on 3.9% of the applicants, the forest's on 2.6%.



Left: test accuracy over 30 random 70/30 splits, single tree against bagging against random forest, the red line the mean. Right: how many verdicts change with another 1,200 loans, bagging against the forest. Reading left to right: the median rises, the box narrows, and the whiskers pull in — mean and spread both improve together, as sections 5.2 and 6 lead us to expect.

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7. Feature importance, and its limits

A random forest reports **feature importance**: how much each column helped the splits. Each split's gain ΔG, weighted by the share of applicants it splits, is credited to the column it asks about; the credits are summed per column over every split of every tree and divided by their total, so that the columns add up to 1. On the depth-2 tree of section 3:

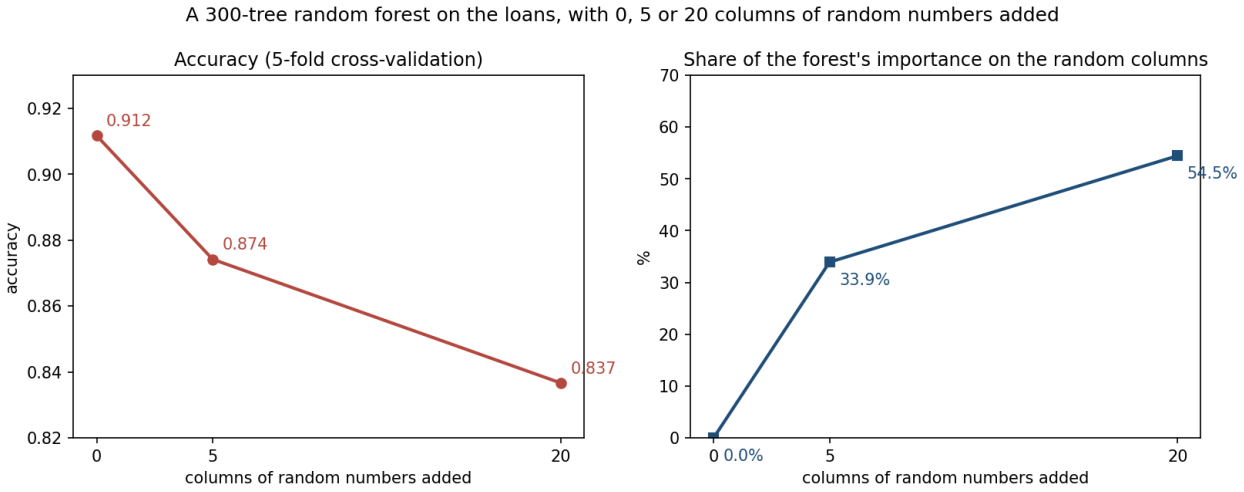
Split	Column	Applicants	Gain ΔG	Weight = applicants ÷ 1,200	Credit = ΔG × weight
root	debt ratio 0.82	1,200	0.1438	1.0000	0.1438
left	income 28	971	0.0836	0.8092	0.0676
right	debt ratio 0.947	229	0.0026	0.1908	0.0005

Debt ratio is credited $0.1438 + 0.0005 = 0.1443$ and income 0.0676, a total of 0.2119; divided by it, debt ratio 0.681 and income 0.319 — what `feature_importances_` returns for that tree. A 300-tree forest on the 1,200 loans gives debt ratio 0.608 and income 0.392. It is tempting to read this as "the forest tells you which features matter" — and with only the two real features in this dataset, it does exactly that.

The question is what happens once most of the columns carry nothing.

Noise columns added	Total columns	Cross-validated accuracy	Importance on noise columns
0	2	0.912	0.0%

Noise columns added	Total columns	Cross-validated accuracy	Importance on noise columns
5	7	0.874	33.9%
20	22	0.837	**54.5%**



Left: cross-validated accuracy falls as columns of random numbers are added. Right: the share of the forest's own importance that lands on those columns rises at the same time. Accuracy degrading gracefully gives no hint that the importance behind it has become largely wrong.

What "more than half" means: the importances of all 22 columns add up to 1, and the 20 random columns together take 0.545. One by one, the real columns are still the top two — in the forest of section 7.1, trained on 70% of the loans, debt ratio 0.319 and income 0.133, against 0.027 for an average random column — but there are twenty of them. The problem is the mass, not the ranking: a reader who did not know they were noise would conclude that they explain more than income and debt together.

More irrelevant columns cost accuracy — the same direction as lesson 6's curse of dimensionality, though the mechanism is different — and, more strikingly, an increasing share of the forest's own importance scores lands on columns that are `np.random.normal` noise by construction, wired to nothing. **With 20 pure-noise columns against 2 real features, 54% of the forest's importance mass sits on noise. A predictable mistake:** "so 54% of the forest's predictions are decided by noise". Reasonable, since a share of importance reads like a share of responsibility. But the 54% adds up credit for splits, the impurity gain each split bought on the training rows; it is not a share of predictions or of accuracy. Section 7.1 measures that second thing, and there the noise columns are worth almost nothing.

Two things hand credit to noise, and they are worth taking in order.

The menu. The first follows from section 6's restriction directly. With `max_features = "sqrt"` and $n = 22$ total columns, each split considers $k = \lfloor\sqrt{22}\rfloor = 4$ candidates. By symmetry — the number of size-k subsets of n features that contain one fixed feature is $\binom{n-1}{k-1}$, against $\binom{n}{k}$ subsets in total — any one specific column, including either real feature, is a candidate with probability

$$\frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n} = \frac{4}{22} \approx 18.2\%$$

The middle equality is worth one line of cancellation, since it is what makes the answer so simple: writing both binomials as factorials,

$$\frac{(n-1)!}{(k-1)!\,(n-k)!} \cdot \frac{k!\,(n-k)!}{n!} = \frac{k!}{(k-1)!} \cdot \frac{(n-1)!}{n!} = \frac{k}{n}$$

— every factorial cancels except one factor at each end.

So any one column is *excluded* from a given split with probability $1 - 4/22 \approx 81.8\%$. For both real features to be left out together, all four candidates must come from the 20 noise columns, which happens with probability

$$\frac{\binom{20}{4}}{\binom{22}{4}} = \frac{4{,}845}{7{,}315} \approx 0.662$$

— about two splits in three offer no real feature at all, and the tree must choose its best option among whichever noise columns happened to be drawn.

Small groups. The menu is not the whole story. Give every split all 22 columns, with `max_features=None`, and the noise share does not fall to zero: it falls from 54.5% to **28%**. The rest comes from the bottom of the trees. A node deep in a fully grown tree holds a handful of applicants, and among twenty columns of random numbers one will almost always separate that handful a little better than income or debt can, purely by chance. Each such gain is small, but it is not zero, and enough of them accumulate across hundreds of trees and thousands of splits to produce the shares measured above.

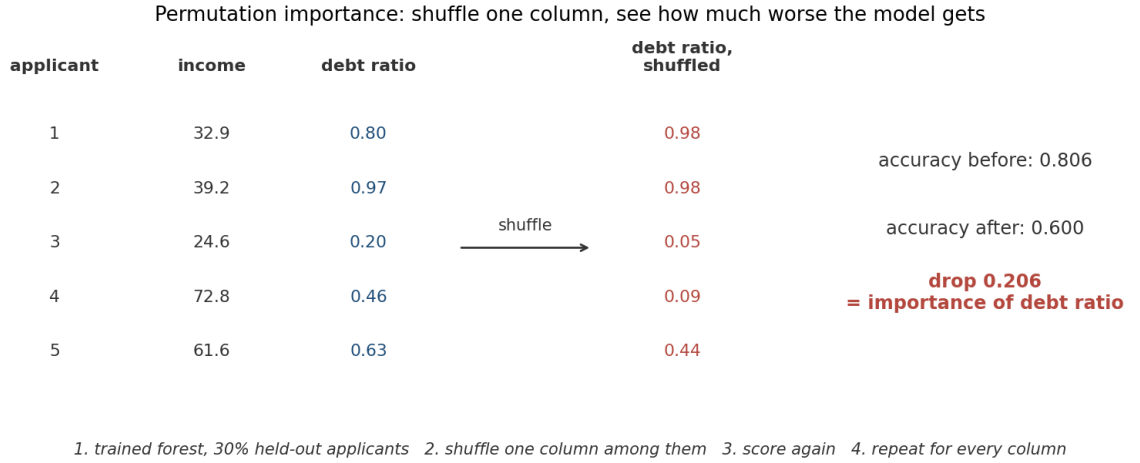
The same run says something about the default, too. With every column on offer, cross-validated accuracy rises from 0.837 to **0.897**: with 20 useless columns out of 22, the random menu costs more accuracy than decorrelating the trees buys back. `"sqrt"` is a good default, not a law, and the number of useless columns is what decides it.

The predictable mistake, and why the instinct is sound. It is entirely reasonable to expect an average over hundreds of trees to wash out noise — and section 5.2's variance-reduction result is real, measured, and points the same way. The averaging genuinely reduces the *variance of the prediction*. It does not make the credit for noise small, because both causes above act inside every tree, and averaging many trees that each credit noise gives an average that credits noise. A second, milder version of the same instinct: "with every feature on offer, noise gets nothing". It is sound near the root, where the real features win easily; it fails at the bottom of the trees, where the groups are small.

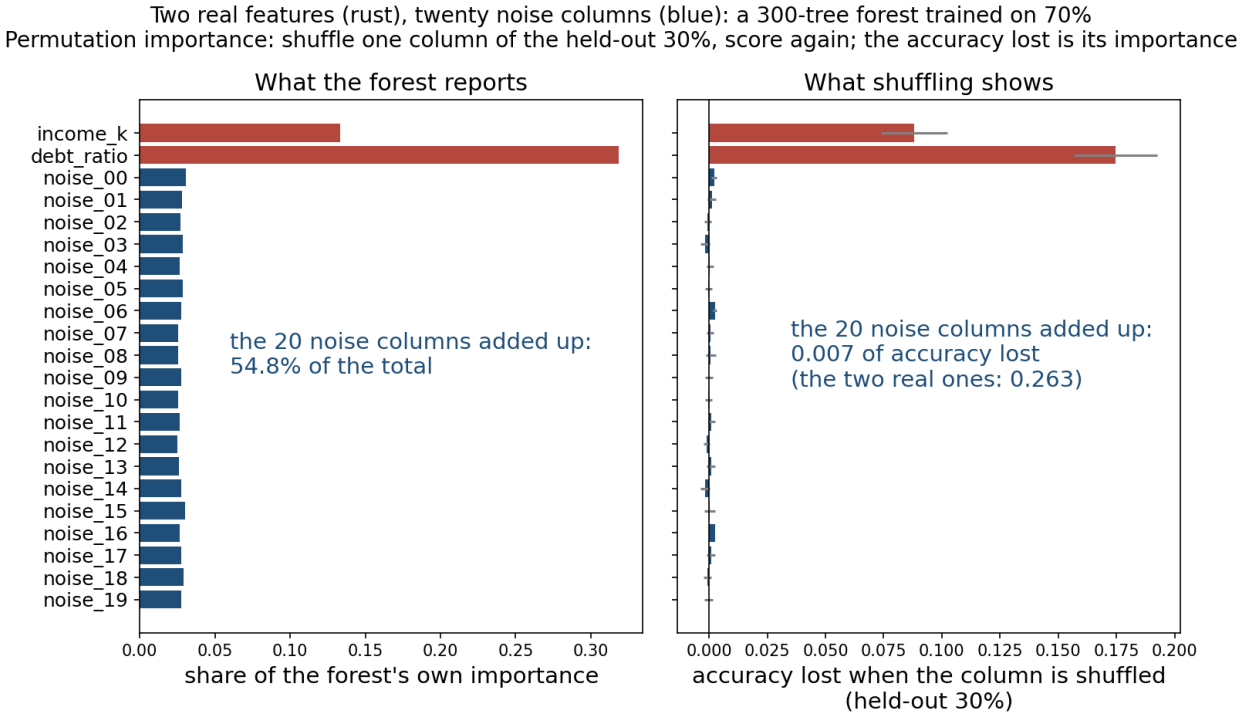
7.1 The check: permutation importance

The importance a forest reports is computed from its own training splits, so it is a claim the forest makes about itself. **Permutation importance** tests the claim from outside. Take the fitted model and data it was not trained on; shuffle the values of one column, so that the column no longer says anything about the applicant in its row; score the model again. The accuracy lost is that column's

importance. A column the model relies on costs accuracy when shuffled; a column it never really used costs almost nothing.



The procedure on five real held-out applicants: the debt-ratio column before and after one shuffle among the 360, and the accuracy before (0.806) and after (0.600); the drop, 0.206, is one estimate of the debt ratio's importance. Notebook 2 repeats each shuffle 20 times and averages: 0.175.



The 20-noise-column forest, fit on 70% of the applicants and checked on the other 30%. Left: the importance the forest reports for itself, more than half of it on the twenty noise columns. Right: the same 22 columns by permutation on the held-out 30%. The two real features carry all of it; every noise column sits at zero, within a few thousandths either way.

Shuffling debt ratio costs 0.18 of accuracy, shuffling income 0.09, and each noise column between 0.003 lost and 0.002 gained. *Almost* nothing, not exactly nothing: the shuffle is random and the

test set finite, so a useless column moves the score by a few thousandths in either direction. And the check has to run on held-out data. On the training rows a fully grown forest has memorised its noise columns along with everything else, and shuffling them costs it accuracy there — the same flaw as the built-in importance, measured a second way.

The practical consequence: with few real features against many candidate columns, treat a random forest's importances as suggestive, not definitive. Check them by permutation on held-out data — one call to `sklearn.inspection.permutation_importance` — and, before asking the forest which columns matter, ask which ones could plausibly matter at all.

> **Try this:** in notebook 2, section 6, run the permutation check on the *training* rows instead of the held-out ones, and measure how much the twenty noise columns cost there, added up, against the 0.007 they cost on held-out data. The paragraph above says which way it moves; the size is yours to find.

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8. Gradient boosting: correcting mistakes in sequence

A note of history: the first boosting algorithm to work in practice was AdaBoost, by Yoav Freund and Robert Schapire (1995–1997), which reweighted the examples the trees so far got wrong; it won them the Gödel Prize in 2003. Jerome Friedman's 2001 paper recast the idea as gradient descent on a loss — the version this section teaches and scikit-learn implements.

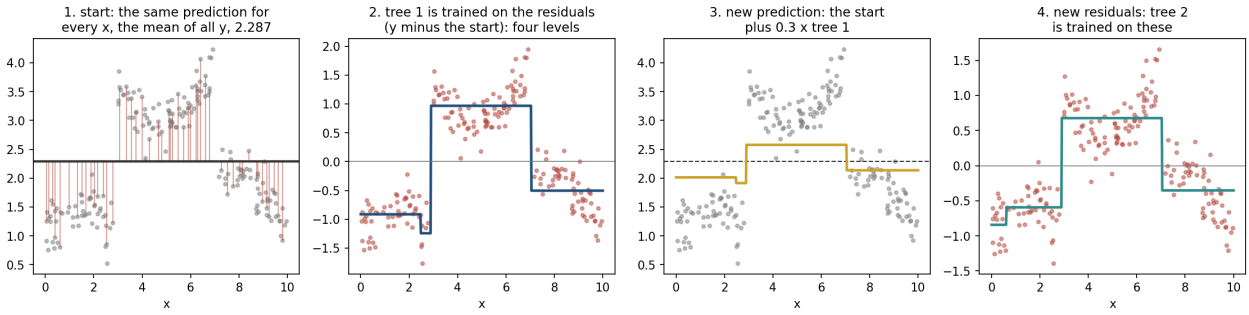
Bagging and random forests build trees **independently** and average at the end. **Boosting** builds trees **in sequence**: each new tree is fit specifically to correct what the trees so far got wrong.

$$F_0(x) = \bar{y}, \qquad F_t(x) = F_{t-1}(x) + \alpha \cdot h_t(x)$$

where h_t is a shallow tree — depth 2 or 3, a **weak learner** — fit to what remains unexplained by F_{t-1}, and α is the **learning rate**: how much of each new tree's correction is actually applied. F_0 is the best single constant for the loss being minimised: for squared error, the case of this section, it is the mean $\bar{y}$; for classification, section 9, it will be the log-odds of the base rate.

The trees are **regression trees**, even when the task is classification, because what they are fit to is a number, not a class. A regression tree is section 2's algorithm with two changes: each leaf predicts the mean of the values that reach it, and a split is scored by how much it lowers the squared error around those means, in place of the Gini impurity. The greedy search over features and thresholds is unchanged.

Boosting, drawn. On the toy data of the worked example below — 200 noisy points around a step plus a sine wave — the first two steps one at a time. The start is the same prediction for every x, the mean of all 200 y, 2.287: before any tree the model cannot tell one x from another, and among all constants the mean has the smallest squared error. Tree 1 is trained on the residuals, y minus the start: it splits x at 2.48, 2.89 and 7.05 and predicts -0.914, -1.243, $+0.967$ and -0.504. The new prediction is the start plus 0.3 times tree 1 — at $x = 5$, $2.287 + 0.3 \times 0.967 = 2.578$. Tree 2 is trained on the new residuals, and so on. The average size of a residual falls from 0.838 to 0.601 after tree 1.



Left to right: the flat start and the residuals; tree 1 trained on them, four levels; the new prediction, the start plus 0.3 times tree 1; the new residuals, which tree 2 is trained on.

A predictable mistake. "Each tree corrects the previous tree." Each new tree corrects what the whole sum so far still gets wrong: $F_1 = F_0 + \alpha h_1$, tree 2 is trained on what F_1 gets wrong, and the final prediction is the start plus α times every correction.

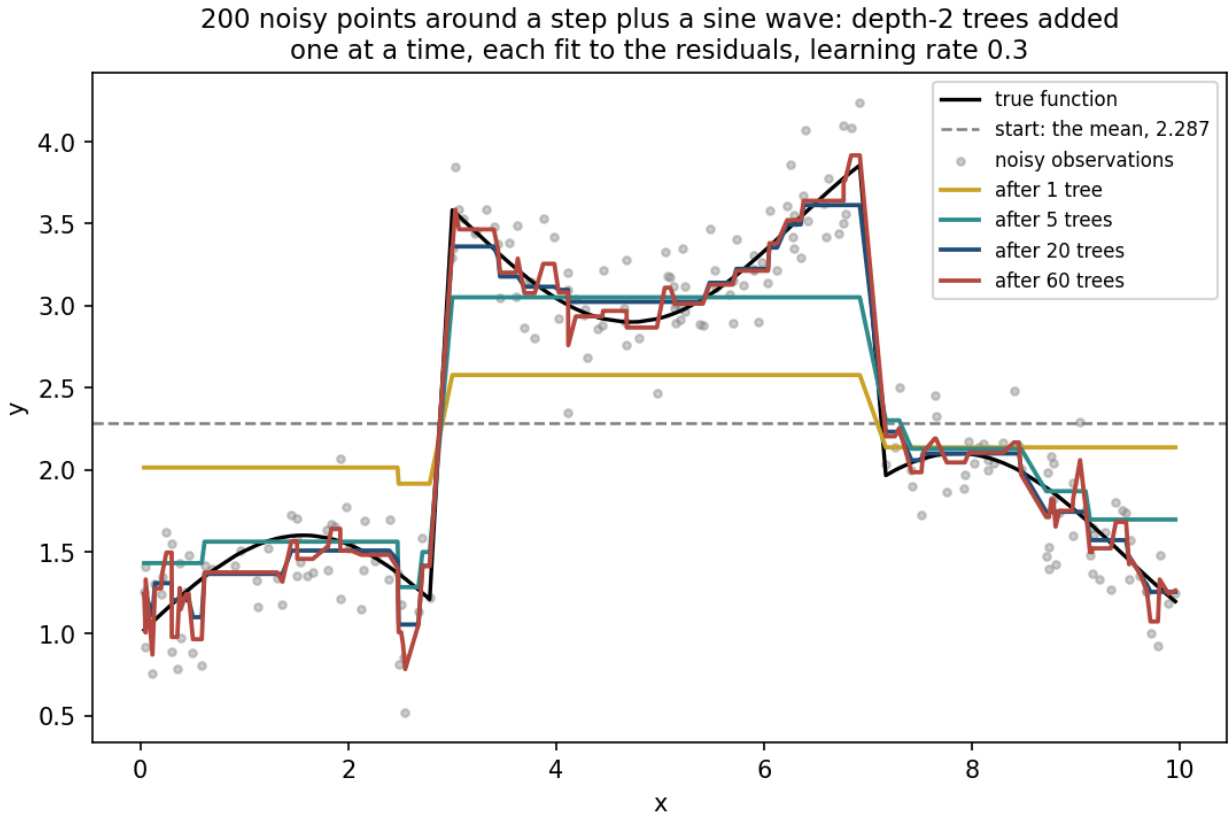
A predictable mistake, and why it is reasonable. "The final task is classification, so each boosted tree votes default or repaid, like a tree in a forest." Every ensemble before this one was made of trees that vote, so the expectation is natural. But a boosted tree never outputs a class. It outputs a correction that is added to a running score F, and only the final score is turned into a class.

For squared-error regression, "what remains unexplained" has an exact name, the **residual** $y - F_{t-1}(x)$. This is a special case of a more general idea — **functional gradient descent**: h_t is fit to approximate the negative gradient of the loss with respect to the prediction F_{t-1} at each training point, the direction in which each prediction should move. It is lesson 3's gradient descent, moving the predictions of the points instead of coefficients, with a tree to draw that move as a function of x. For $L(y, F) = \frac{1}{2}(y - F)^2$,

$$-\frac{\partial L}{\partial F} = -(F - y) = y - F$$

so "fit the residual" and "fit the negative gradient" coincide exactly for squared error, and the general recipe reduces to the familiar one.

Worked example. A synthetic target — a three-level step function plus a gentle sine wave, observed with noise — was fit by 60 depth-2 trees at $\alpha = 0.3$:



A sum of shallow trees converging on the residuals; the dashed grey line is the start, the mean 2.287. One tree barely moves it — the gold curve is the start plus 0.3 times tree 1: $2.287 - 0.274 = 2.01$ *on the left,* $2.287 + 0.290 = 2.58$ *in the middle,* $2.287 - 0.151 = 2.14$ *on the right (mean squared error, MSE, against the true function: 0.395). Five trees sketch the coarse shape (MSE 0.068). Twenty trace the step function closely, oscillation included (MSE 0.012) — the best of the four shown; counted tree by tree, the lowest is at 19 trees, 0.0116 against 0.0119 at 20. By sixty trees the ensemble is tracking individual noisy observations as well as the underlying shape, and the error against the (noise-free) truth has risen back to 0.018: the same overfitting story as section 3's unconstrained tree, reached by a different route — not one tree grown too deep, but too many shallow trees each chasing whatever residual noise is left.*

The error on the training points tells a different story from the error against the truth:

Trees	MSE on the 200 training points	MSE against the true function
1	0.476	0.395
5	0.119	0.068
20	0.046	**0.012**
60	0.025	0.018

The training error keeps falling; the error against the truth turns up after 20 trees. The noise added to the points has variance $0.25^2 = 0.0625$, and from 20 trees on the training error is below it — a sign that the model has started to learn the noise itself, though not a proof, since the 200 noise

values have a sample variance of their own. The proof is the last column: the error against the true function, which the model never sees, turns up after 20 trees. With real data the truth is unknown, and section 10 puts cross-validation in its place.

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9. Boosting for classification: the same idea, a different gradient

Classification does not have a "residual" in the same literal sense, but the functional-gradient recipe still applies — only the loss changes. With $y \in \{0, 1\}$ and the ensemble's output converted to a probability by the logistic function, $p = \sigma(F) = 1/(1 + e^{-F})$, the loss is the **log-loss** lesson 4 used for logistic regression:

$$L(y, F) = -[y \log p + (1 - y) \log(1 - p)]$$

By the chain rule, $\dfrac{\partial L}{\partial p} = -\dfrac{y}{p} + \dfrac{1 - y}{1 - p}$ and $\dfrac{\partial p}{\partial F} = p(1 - p)$ (the logistic function's own derivative), so

$$\frac{\partial L}{\partial F} = \frac{\partial L}{\partial p} \cdot \frac{\partial p}{\partial F} = \left(-\frac{y}{p} + \frac{1 - y}{1 - p} \right) p(1 - p) = p - y$$

and the negative gradient — the **pseudo-residual** each tree is fit to — is $y - p$: the true label minus the current predicted probability, exactly the error term lesson 4's logistic regression gradient descent moved against. Boosting for classification is fitting a sequence of small trees to that same quantity, one correction at a time, rather than adjusting a fixed set of linear coefficients against it in one continuous descent.

Where it starts, and the first residuals. The starting score is the best constant, the log-odds of the base rate: 464 of the 1,200 applicants defaulted, $p = 0.387$, so $F_0 = \log(464/736) = -0.461$, which the logistic function turns back into 0.387 for every applicant. The first tree is then fit to $y - p$: **+0.613** for every applicant who defaulted and **−0.387** for every one who repaid. Where that tree finds a group of mostly defaulters, it raises their score; each later tree works on what the earlier ones left.

A predictable mistake, and why it is reasonable. "So the update is $p + \alpha(y - p)$." The pseudo-residual is a difference of probabilities, so adding it to a probability looks natural. But the trees correct the score F, not p: F moves by $\alpha h_t(x)$, and only then does the logistic function turn the new F into a new p. That is what keeps p between 0 and 1 however large the corrections add up.

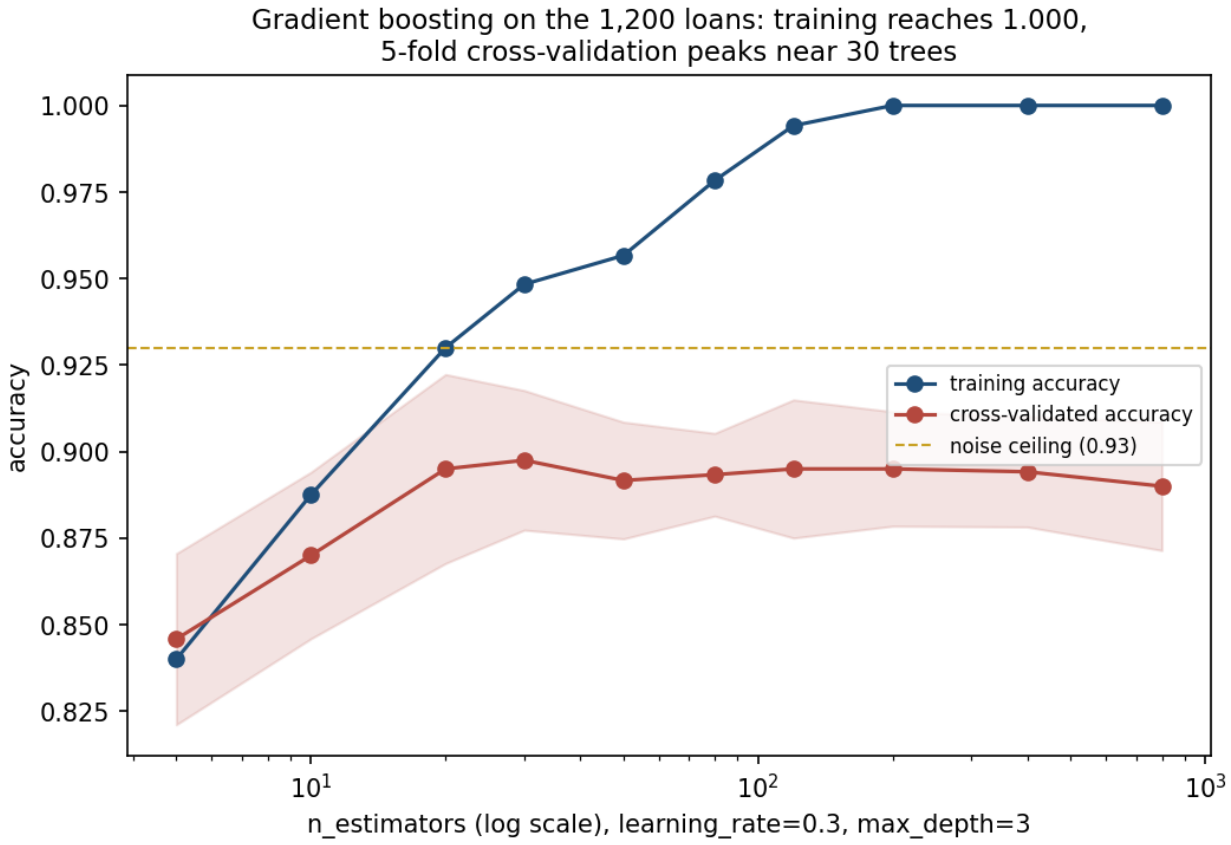
One detail of scikit-learn's implementation, for anyone who reads its source. The splits of each tree are chosen on $y - p$ exactly as described; the value written in each leaf is then recomputed as $\sum(y - p) \,/\, \sum p(1 - p)$ over the leaf's examples — one Newton step on the log-loss in place of the plain mean. The splits come from the residuals, the size of each step from the curvature of the loss. Nothing in this lesson depends on it.

A default `GradientBoostingClassifier` — 100 trees, learning rate 0.1 — reaches **0.902 ± 0.020** cross-validated accuracy on the loan data, already ahead of any single tree. **A predictable mistake**, once section 10 is read: "30 trees beat the default 100". Reasonable, since 30 is the peak there and 100 is the default here. But section 10 uses a learning rate of 0.3 and the default uses 0.1: two different configurations, not one knob turned.

10. Overfitting in boosting: two knobs that trade off

Unlike bagging, boosting keeps reducing its training loss for as long as it runs, because each tree is built specifically to reduce what remains. With nothing to stop it, that includes the 7% of labels that are noise by construction. The loss is the quantity that falls at every step; training accuracy follows it on the whole but not at every single tree — fit to all 1,200 applicants with 800 trees, it dips at a handful of steps, too few to show in the table.

`n_estimators` ($\alpha = 0.3$)	Training accuracy	Cross-validated accuracy
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.897 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019



Training accuracy reaches 1.000 by 200 trees and stays there — the same ceiling section 3's unconstrained tree reached in one step, here approached gradually. The gold dashed line is the noise ceiling: with 7% of the labels flipped, no honest model can expect much above about 0.93 on new applicants, so a training accuracy of 1.000 means the flipped labels have been learned by heart. Cross-validated

accuracy climbs quickly, peaks around 30 trees, and drifts down by roughly a point over the following 770 trees.

The overfitting is real but **gentler** than a single unconstrained tree's: that tree lost 3 points of cross-validated accuracy past its peak (section 3); 800 boosted trees lose about 1, even though training accuracy has been at 1.000 since well before that point. That gentleness is why the number of trees has to be chosen on held-out data rather than left to run — the damage per added tree is small and easy to miss on a single run, and it is still there. There are two ways to choose it. This lesson cross-validated a list of tree counts and kept the best. The other is **early stopping**: set aside part of the training data, watch the score on it as trees are added, and stop once it has not improved for a fixed number of rounds; in scikit-learn it is the `n_iter_no_change` argument of `GradientBoostingClassifier`.

Learning rate and tree count are not independent: both control how much total correction the ensemble applies, so they trade off against each other. Fixing `n_estimators = 120` and varying only α:

Learning rate α	Training accuracy	Cross-validated accuracy
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

A learning rate of 0.02 has not finished fitting the signal in 120 trees — training accuracy is still under 0.89. A learning rate of 1.0 has memorised the training set completely (1.000) and given back more than two points of cross-validated accuracy for it, 0.902 down to 0.878. The best setting sits in between: large enough to make real progress within the tree budget, small enough to leave each individual correction easy for later trees to undo if it turns out to be wrong.

A predictable mistake, and why it is reasonable. "Only α times the number of trees matters, so half the learning rate with twice the trees gives the same model." The paragraph above has just said that both knobs set the total correction, so the shortcut is natural. But they are not interchangeable: each tree is fit to what the ensemble before it still gets wrong, so many small steps follow a different path from a few large ones. A large step overshoots on the noisy labels and the next trees spend themselves undoing it; a small one lets later trees correct an early mistake. The trade-off is real; the equivalence is not.

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11. Choosing among trees, forests and boosting

Every method from this lesson, on the same five folds of the 1,200 loans; the third column is the standard deviation of the five fold scores:

Model	Cross-validated accuracy	± over the 5 folds
Majority baseline	0.613	
Logistic regression	0.748	0.030

Model	Cross-validated accuracy	± over the 5 folds
Single tree, unconstrained	0.852	0.021
Single tree, `max_depth = 8`	0.882	0.010
Gradient boosting, 30 trees	0.897	0.020
Single tree, `max_leaf_nodes = 9`	0.901	0.020
Bagging, 100 trees	0.904	0.020
Random forest, 100 trees	**0.911**	0.018

Noise ceiling: 0.93 (1 − the 7% deliberately flipped labels), what the rule that generated the data scores on average. No method reaches it here, and none should be expected to: a flipped label is a coin toss no model can foresee. On a finite test set a score can land a little above it by luck; far above it would point to a bug.

Three findings, read together, are this lesson's real content. First, on this dataset the ensembles — bagging, the random forest, gradient boosting — land within about a point and a half of each other, the top four rows within one spread of each other. **A predictable mistake** is to call gradient boosting the winner: it is presented last, and on tabular data in practice it often wins; here the random forest is on top and boosting at 30 trees is fourth — the most sophisticated method is not automatically the best. Second, a single tree sits among them: the nine-leaf tree of section 4 reaches 0.901, above gradient boosting and a point behind bagging, and a person can read it start to finish. On data whose rule is a handful of axis-aligned rectangles, a small tree is nearly enough; what the ensembles buy here is less a higher score than a steadier one. On the same 30 random splits as section 5.2, the nine-leaf tree averages 0.897 with a standard deviation of 0.020, the random forest 0.908 with 0.0137; section 4 showed where a single tree's verdicts change. Third, the unconstrained tree is the worst tree on this list despite being, in a narrow sense, the most flexible one: flexibility without a stopping rule is not an advantage, a point this course has now made with several different methods.

Situation	Reach for
The decision must be explained to the person it affects	A single tree, small enough to read; if even that is too opaque, none of these models
Tabular data, accuracy matters most, no explanation required	A random forest: the best score here, at its default settings
Accuracy matters, and there is time to tune the number of trees and the learning rate together	Gradient boosting, tuned on validation data
Many columns, few of them informative	Any tree method, checking its importances by permutation (section 7.1): section 7's 54% came with 1,200 rows
Too few rows to set a validation set aside	A random forest, judged by its out-of-bag score

One caveat, in lesson 5's terms. Depth 8 and 30 boosted trees were chosen because they scored best on the same five folds as the leaderboard, so those two leaderboard rows are slightly optimistic; nested cross-validation would remove the bias. Bagging and the random forest ran at their defaults, chose nothing on these folds, and are not affected — and they are still the two best rows. The nine-leaf budget was chosen for readability, not on these folds.

Summary

- A decision tree splits greedily on Gini impurity, $G = 1 - \sum_c p_c^2$, one feature and threshold at a time — verified here against scikit-learn's implementation, prediction by prediction.
- Depth is the bias-variance dial. Cross-validated accuracy peaked at **depth 8** (0.882); an unconstrained tree reached training accuracy **1.000** and lost 3 points of cross-validated accuracy for it.
- A tree's strongest split survives resampling almost unchanged; its weaker splits do not: two trees grown on 65% samples gave opposite verdicts to 140 of the 1,200 applicants (11.7%), three in four in or next to the islands. A budget of nine leaves (`max_leaf_nodes`) cross-validated at 0.901, above depth 8.
- Bagging averages trees fit to bootstrap resamples. Each sample leaves about **36.8%** of rows out-of-bag — the standard limit $(1 - 1/m)^m \to e^{-1}$ — which is why the OOB score is free validation: 300 bagged trees scored 0.9083 out of bag against 0.9025 ± 0.021 cross-validated. The vote cancels only the trees' disagreements: of 29 flipped labels in a test set it got 1 right.
- Averaging stops at what the trees share: a floor of $\rho\sigma^2$, not zero. Over 30 resampled splits bagging got about 17 more of 360 test applicants right than one tree (0.902 against 0.856). A random forest makes the trees less alike by offering each split a random subset of features: 0.908, with the test-accuracy spread going from 0.0185 (single tree) to 0.0153 (bagging) to 0.0137 (random forest). Averaging cannot repair an error every tree shares.
- **With 20 pure-noise columns against 2 real features, 54% of a random forest's feature importance landed on noise** — the number worth carrying out of this lesson. Two causes: `max_features="sqrt"` offers no real feature at two splits in three, and deep splits on small groups credit noise by chance even when every feature is on offer. Permutation importance on held-out data puts the noise columns back at zero.
- Gradient boosting fits shallow trees in sequence to the negative gradient of the loss — the residual for squared error, $y - p$ for log-loss, the same quantity lesson 4's logistic regression descended against. Cross-validated accuracy peaked near 30 trees and drifted down by about a point over the next 770, a gentler overfit than a single unconstrained tree's but not an absent one.
- On this dataset the ensembles landed within about a point and a half of each other, the random forest on top at 0.911 — and the nine-leaf tree came within a point of bagging, because the rule is made of rectangles.

Homework

`Exercises/07_trees_and_ensembles.md`, discussed at the start of Lesson 8, **Friday 4 December 2026**.

Notation used in this lesson

Symbol	Meaning
G	Gini impurity of a group of examples
ΔG	impurity reduction a candidate split buys
p_c	fraction of a group belonging to class c
d	a tree's depth
m, n	number of examples, number of features

Symbol	Meaning
m_g; m_P, m_L, m_R	examples in a group; in a parent and its two children
σ^2, ρ	variance of one tree's prediction, correlation between two trees
k	features offered as candidates at each split (`max_features`)
B	number of trees averaged
F_t, h_t	the ensemble's prediction after t rounds, and the t-th tree added to it
α	**learning rate**, boosting's shrinkage factor — the same symbol as lessons 3 and 9
p	the model's predicted probability, $\sigma(F)$

Further reading

Resource	Type	Why read it
scikit-learn: decision trees	Official docs	The splitting criteria and stopping parameters stated precisely
scikit-learn: ensemble methods	Official docs	Bagging, random forests and boosting side by side, with the parameters that matter
Breiman, *Random Forests* (2001)	Paper	The original derivation of the $\rho\sigma^2$ variance bound used in section 5.2
Friedman, *Greedy Function Approximation: A Gradient Boosting Machine* (2001)	Paper	The functional-gradient view of boosting, section 8's general recipe, from its source
Hastie, Tibshirani & Friedman, *Elements of Statistical Learning*, ch. 9–10, 15	Book	Trees (ch. 9), boosting (ch. 10) and random forests (ch. 15), the definitive treatment
Molnar, *Interpretable Machine Learning*, ch. 5–6	Book, free online	Permutation importance and its failure modes — the fix section 7 points towards